The $qx + 1$ Generalization of Tao's Syracuse Measure: An Endogeny Barrier and Complementary Viewpoints on the Arithmetic Obstruction

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Abstract

We study a $qx + 1$ generalization of Tao's Syracuse random variable [1], the probabilistic object behind the strongest unconditional partial result toward the Collatz conjecture.

A companion paper proves a closed-form annealed pressure equation, $q^{\alpha-1} = 2^{\alpha} - 1$, for the reverse tree of the accelerated $qx + 1$ map, via an exact fibre-counting bijection [8]. Its two roots $\alpha_-(q) < \alpha_+(q)$ satisfy $\alpha_-(q) = 1$ at $q = 3$, reproducing the Growth Exponent Conjecture of Kontorovich–Lagarias and Applegate–Lagarias as this case of a single closed-form equation, and $\alpha_-(q)$ falls below 1 for every $q \geq 5$: a structural transition, sharp rather than gradual. $\alpha_-(q)$ is always unfrozen, in the sense of directed polymers in a random environment, so the identity transfers rigorously to the matching i.i.d. branching-random-walk model, and, as empirical evidence rather than proof, to the arithmetic tree itself. The other root, always frozen, equals 2 at $q = 3$ and separately matches earlier tail-exponent measurements on the classical Collatz reverse tree.

We prove that the tree's fixed-point recursion admits an *endogeny barrier*, in the sense of Aldous–Bandyopadhyay [12]: solutions of the form $G = W \cdot Y$, for Y any invariant factor along the tree bounded away from zero and infinity, solve the same recursion as the canonical solution W and match it in pressure, tail index, and regression slope against the limiting measure. The recursion together with the statistics of the 3-adic digits cannot by itself force this gap to vanish. A three-arm control experiment bounds the residual empirical coupling between sibling subtrees at $\rho_{\text{eff}} \lesssim 0.06$ (95% CI).

Closing the gap connects several open arithmetic statements. We test Wirsching's 1998 Weak Covering Conjecture directly by computation through level 20, and prove an unconditional bound $\beta_{\text{eff}} \leq 1.882712$ on the related cost in Tao's $\beta = 1$ conjecture, from an inter-level cascade identity on the worst-cylinder mass, improving the previous bound of 2.306270. A large-deviation argument shows that even ideal multiplicity inside the conjecture's own cost window falls short of what $\beta = 1$ needs. A companion paper proves Wirsching's first 2003 conjecture on the base-3 Fabius function, treats the second with a corrected reading of its actual target, and tests the third to certified numerical error [10]. Two candidate lemmas from Tao's bivariate Fourier-decay technique each fail to close the gap, for reasons we make exact. One yields a closed-form structure theorem, tested against sustained finite-level growth through level 17. The other, an exact Jensen identity, quantifies a slack factor of 1.88 in one annealed Fourier budget, a slack we show persists across the entire wider family of annealed ℓ^r budgets. A diagonalization in discrete-logarithm coordinates separately proves an unconditional collision growth rate $K_{q,\ell} \geq (q/3)^{\ell}$ for every odd $q \geq 3$, with no lifting hypothesis, reducing the remaining critical case $q = 3$ to a single affine shifted collision.

A companion paper separates Kontorovich and Lagarias's counting-exponent prediction for $5x + 1$ from a competing one due to Volkov, once a raw estimator's own bias is calibrated out

with matched synthetic controls [9]. A directed literature search situates the barrier against the general theory of Fourier decay for self-similar and self-conformal measures [21, 22], and against a recent fixed-modulus orbit statistic [27]; neither supplies a shortcut.

Keywords: Collatz conjecture; $qx + 1$ map; Syracuse random variable; pressure equation; freezing transition; branching random walk; martingale tail index; endogeny; Weak Covering Conjecture; Fabius function; Fourier decay; counting exponent.

Contents

1 Introduction

1.1 The Collatz conjecture and the Syracuse random variable

The Collatz (or $3x + 1$) conjecture asserts that iterating the map

$$C(n) = \begin{cases} n/2 & n \text{ even} \\ 3n + 1 & n \text{ odd} \end{cases}$$

started at any positive integer n eventually reaches 1. Despite its elementary statement, the conjecture remains open, and the strongest unconditional result toward it is due to Tao [1], who showed that *almost all* orbits (in the sense of logarithmic density) attain almost bounded values. Tao's proof introduces the *Syracuse random variable*: fixing n, let $a_1, a_2, \dots, a_n$ be i.i.d. geometric random variables with $\Pr(a_i = k) = 2^{-k}$ for $k \geq 1$, and define

$$F_n(a) := \sum_{j=1}^{n} 3^{j-1}\, 2^{-a_{[1,j]}} \pmod{3^n}, \qquad a_{[1,j]} := a_1 + \cdots + a_j, \tag{1}$$

where 2^{-1} denotes the multiplicative inverse of 2 modulo 3^n. The random variable $\mathrm{Syrac}(\mathbb{Z}/3^n\mathbb{Z}) := F_n(a_1, \dots, a_n)$ models the residue of an accelerated Collatz orbit modulo 3^n after n odd steps, and Tao's proof reduces the conjecture's target statement to controlling the distribution of this random variable, in particular via a Fourier decay estimate:

Proposition 1.1 (Tao 2022, Prop. 1.17)**.** *For every $A > 0$ there is C_A such that for all $n \geq 1$ and every $\xi \in \mathbb{Z}/3^n\mathbb{Z}$ with $3 \nmid \xi$,*

$$\left|\mathbb{E}\, e\big(\xi\, F_n(a)/3^n\big)\right| \leq C_A\, n^{-A},$$

with C_A uniform in n and ξ (subject only to $3 \nmid \xi$).

This is a super-polynomial (though not exponential) decay estimate, and it is the technical heart of Tao's theorem. A recursive consequence of the construction (1), which we use repeatedly below, is the following *exact self-similarity* identity, verified directly against Tao's text:

Proposition 1.2 (Tao 2022, eq. (1.23))**.** *For $k \le n$,*

$$\mathrm{Syrac}(\mathbb{Z}/3^n\mathbb{Z}) \bmod 3^k \;\equiv\; \mathrm{Syrac}(\mathbb{Z}/3^k\mathbb{Z})$$

as an identity in law. Equivalently, writing $F_n(a) = 2^{-a_1}\big(1 + 3\,F_{n-1}(a_2,\dots,a_n)\big) \bmod 3^n$, the coarsest k coordinates of $\mathrm{Syrac}(\mathbb{Z}/3^n\mathbb{Z})$ *depend only on $(a_1,\dots,a_k)$, the digits* adjacent to the root *in the indexing of* (1)*, rather than on the deepest ones.*

The orientation convention embedded in Proposition 1.2 (that the coordinate adjacent to the root is a_1 rather than a_n) is a recurring source of error when translating Tao's construction into reverse-tree language.

1.2 This paper's contributions

Building on [1] and on the classical predecessor-density program of Wirsching [4, 5], the paper proceeds as follows. Section 3 restates the closed-form pressure equation for the accelerated $qx+1$ tree and its structural transition at $q = 5$, proved in full in a companion paper [8]. Section 4 proves the endogeny barrier: an explicit family of non-endogenous solutions to the tree's fixed-point recursion, indistinguishable from the canonical one by pressure, tail index, or regression slope against the limiting measure. Section 5 bounds the residual empirical coupling between sibling subtrees at $\rho_{\mathrm{eff}} \lesssim 0.06$. Sections 7 and 9 test Wirsching's Weak Covering Conjecture by direct computation through level 20 and prove the unconditional bound $\beta_{\mathrm{eff}} \le 1.882712$ on the related cost in Tao's $\beta = 1$ conjecture. A companion paper proves Wirsching's first 2003 conjecture, treats the second, and tests the third numerically, all restated without proof in §9 [10]. Section 6 restates a companion paper's calibrated separation of the Kontorovich–Lagarias and Volkov counting-exponent predictions for $5x+1$ [9]. Section 10 executes two candidate routes to Regime 3 of §8 to completion, proving along the way an unconditional collision-growth bound $K_{q,\ell} \ge (q/3)^\ell$ for every odd $q \ge 3$ (Theorem 10.15). Section 11 situates the barrier against the general theory of Fourier decay for self-similar measures and against a recent fixed-modulus orbit statistic [27]. The eight resulting open points, (O1) through (O8), and how progress on one would transfer to several others, are collected in §13.

2 Setup: the accelerated $qx + 1$ map and its reverse tree

Fix an odd integer $q \ge 3$ coprime to 2. The *accelerated $qx+1$ map* is

$$T_q(n) = \frac{qn+1}{2^{v_2(qn+1)}}, \qquad n \text{ odd},$$

where v_2 is the 2-adic valuation. For $q = 3$ this is the standard accelerated Collatz map. The *reverse tree* rooted at an odd integer u coprime to q is built by, at each odd node u, adjoining a child $w_a := (2^a u - 1)/q$ for every $a \ge 1$ such that $2^a u \equiv 1 \pmod q$; such w_a is automatically an odd integer whenever it is an integer, since an odd numerator divided evenly by an odd denominator yields an odd quotient. Writing $d := \mathrm{ord}_q(2)$ for the multiplicative order of 2 modulo q, admissibility of a for a given parent u depends only on $u \bmod q$: *if* some $a_0(u) \in \{1,\dots,d\}$ satisfies $2^{a_0(u)}u \equiv 1 \pmod q$, it is unique, and the admissible exponents are exactly $a \equiv a_0(u) \pmod d$. Such $a_0(u)$ exists exactly when $u \bmod q$ lies in the subgroup $\langle 2\rangle \le (\mathbb{Z}/q\mathbb{Z})^\times$ generated by 2; nodes with $u \equiv 0 \pmod q$, and, more generally, nodes whose residue mod q falls outside $\langle 2\rangle$, are *sterile* (no admissible children). When 2 is a primitive root mod q (as for $q = 3, 5, 9$, all used numerically below), $\langle 2\rangle$ is all of $(\mathbb{Z}/q\mathbb{Z})^\times$ and $u \equiv 0 \pmod q$ is the only sterile class; for $q = 3$ this recovers the familiar fact that multiples of 3 have trivial reverse subtrees. In general, however, $d = \mathrm{ord}_q(2)$ can be a proper divisor of $\varphi(q)$: at $q = 7$, $d = 3 < \varphi(7) = 6$ and $\langle 2 \rangle = \{1, 2, 4\}$, so $u \equiv 3, 5, 6 \pmod 7$ are sterile too, beyond $u \equiv 0$. This does not affect the pressure identity of Theorem 3.1 below (whose proof, given in full in a companion paper [8], works in the reverse direction, from an unconstrained exponent sequence to its unique root, and never invokes existence of $a_0(u)$ for every u), nor the numerical enumeration reported in §6 below, whose companion-paper implementation [9] already excludes exactly these classes; it is a correction to this structural description only.

We count distinct integer labels rather than path occurrences. Define

$$N_u(x) := \#\{n \le x : n \text{ odd and } T_q^k(n) = u \text{ for some } k \ge 0\}.$$

This convention removes repetitions caused by cycles. In particular, $N_u(x) \le \lceil x/2 \rceil$.

On this tree we study the linear population functional G satisfying the fixed-point recursion

$$G(u) = \sum_{a \equiv a_0(u)\ (d)} q \cdot 2^{-a}\, G(w_a), \tag{2}$$

which for $q = 3$ is exactly the martingale-type recursion underlying the $G(v)/\mu$ line of investigation summarized in §4 below, and which is the direct reverse-tree analogue of (1).

3 The closed-form pressure equation

This section summarizes, without proof, the pressure-equation results a companion paper establishes in full [8]; the rest of this paper depends on the objects and facts stated here.

On the reverse tree of §2, the population functional G of (2) has an exact annealed pressure identity. For a root residue $u_0 \bmod q^k$, the depth-k partition function is $Z_k(\alpha; u_0) := \sum (q \cdot 2^{-a_1})^\alpha \cdots (q \cdot 2^{-a_k})^\alpha$, summed over admissible exponent sequences $(a_1, \ldots, a_k)$ rooted at u_0 (§2); *tilting* the geometric branch law by α means replacing $\Pr(a) = 2^{-a}$ with $p_\alpha(a) := q^{\alpha-1} 2^{-\alpha a}$, which sums to 1 exactly at a root of the pressure equation below, and the *tilted measure* μ_α is the law of the resulting weighted digit sum.

Theorem 3.1 (Exact annealed pressure identity)**.** *For every odd $q \ge 3$ and every $k \ge 1$, the average of $Z_k(\alpha; \cdot)$ over all q^k root residues equals exactly $\big(q^{\alpha-1}/(2^\alpha - 1)\big)^k$; the associated annealed pressure is*

$$\rho_{\mathrm{ann}}(\alpha) = \frac{q^{\alpha-1}}{2^\alpha - 1}, \tag{3}$$

giving the pressure equation at $\rho_{\mathrm{ann}} = 1$:

$$q^{\alpha-1} = 2^\alpha - 1. \tag{4}$$

The function $\alpha \mapsto q^{\alpha-1}/(2^\alpha - 1)$ is log-convex, so (4) has at most two positive roots; $\alpha = 1$ is always one of them (mass conservation). Write $\alpha_-(q) < \alpha_+(q)$ for the two roots when both exist. At $q = 3$, $\{\alpha_-, \alpha_+\} = \{1, 2\}$; for every odd $q \ge 5$ the second root falls strictly below 1, so $\{\alpha_-, \alpha_+\} = \{\alpha^*(q), 1\}$ with $\alpha^*(5) = 0.650919$, $\alpha^*(7) = 0.373501$, $\alpha^*(9) = 0.258108$: a sharp structural transition at $q = 5$. At $q = 3$, $\alpha_-(q) = 1$, and the pressure equation reproduces the Growth Exponent Conjecture of Kontorovich–Lagarias [11] and Applegate–Lagarias [18, 19], open since 1995, as this case of a single closed-form equation across every odd q (Conjecture 3.5 below). The other root, $\alpha_+ = 2$ at $q = 3$, separately matches the tail exponent measured on the classical

Collatz reverse tree in earlier, classical-map-only work of this project, by a Hill estimator and, independently, by extreme-value theory (not detailed in this paper).

The two roots sit on opposite sides of a quenched/annealed freezing transition, in the sense of directed polymers in a random environment: writing $P(\alpha) := \log \rho_{\mathrm{ann}}(\alpha)$ and $s(\alpha) := P(\alpha) - \alpha P'(\alpha)$ for the associated entropy, the following holds for every odd $q \geq 3$.

Proposition 3.2 (The freezing transition is unconditional)**.** *For every odd $q \geq 3$: $s(\alpha_-) > 0$ (*unfrozen*) and $s(\alpha_+) < 0$ (*frozen*).*

In the unfrozen phase, quenched and annealed growth coincide for the matching i.i.d. branching-random-walk model [16], so the pressure identity transfers rigorously to that model's counting exponent.

Theorem 3.3 (Structural transition: exponent, i.i.d. model)**.** *Let $N(x)$ be the population counting function of the i.i.d. branching-random-walk model whose annealed pressure is exactly $\rho_{\mathrm{ann}}(\alpha) = q^{\alpha-1}/(2^\alpha - 1)$. Since $\alpha_-(q)$ is always unfrozen, $\log N(x)/\log x \to \alpha_-(q)$ almost surely, for every odd $q \geq 3$.*

Proposition 3.4 (Finer asymptotic, conditional)**.** *Conditional on a renewal-type refinement (spine decomposition and exponential tilting for the two-sided random walk underlying the model, not yet carried out in this exact form; see the companion paper for the gap in the literature), the counting function has the sharper asymptotic $N(x) \sim W \cdot x^{\alpha_-(q)}$ almost surely and uniformly, including at $q = 3$, for an almost surely positive, non-degenerate random scale factor W.*

Conjecture 3.5 (Structural transition: exponent, arithmetic tree)**.** *For every admissible root u, the actual reverse tree of T_q satisfies $N_u(x) = x^{\alpha_-(q)+o(1)}$: exponent 1 for $q = 3$ (the classical Growth Exponent Conjecture, open since 1995, best unconditional partial result $\pi_a(x) \geq x^{0.84}$ [15]), strictly below 1 for every odd $q \geq 5$. The pressure identity reproduces this as a special case rather than resolving it; no analogous quenched-equals-annealed theorem is known for the arithmetic recursion itself, only for the matching stochastic model. A calibrated empirical test at $q = 5$ against a competing prediction is in a companion paper [9].*

The pressure root also governs a q-adic density martingale and an L^p collision criterion.

Theorem 3.6 (q-adic density martingale, absolute continuity criterion)**.** *Let α be either root of (4) and U Haar-uniform on $\mathbb{Z}_q$. Then $M_k^{(\alpha)}(U) := Z_k(\alpha; U \bmod q^k)$ is a nonnegative mean-one martingale converging almost surely to a limit $W_{q,\alpha}$, identified with the density of the absolutely continuous part of the tilted projective measure μ_α relative to Haar measure. At the frozen root $\alpha_+(q)$, $W_{q,\alpha_+} = 0$ almost surely and μ_{α_+} is singular.*

Theorem 3.7 (L^p collision criterion)**.** *Let $F_k := \sum_{i=1}^k q^{i-1} 2^{-S_i} \bmod q^k$, $S_i := a_1 + \cdots + a_i$, for a_i i.i.d. under the tilted law p_α; the fibre bijection underlying Theorem 3.1 gives $Z_k(\alpha; u) = q^k \Pr(F_k = u)$, so F_k's law $\mu_{\alpha,k}$ modulo q^k and $M_k^{(\alpha)} = q^k \mu_{\alpha,k}$ agree with Theorem 3.6's definition. The projective measure μ_α has a density in $L^p(\mathbb{Z}_q)$ if and only if*

$$\|M_k^{(\alpha)}\|_p^p = q^{k(p-1)} \sum_u \mu_{\alpha,k}(u)^p \tag{5}$$

is bounded uniformly in k. For integer p, this sum is the probability that p independent tilted Syracuse sums agree modulo q^k; at $q = 3$, $\alpha = 1$, $p = 2$ it is exactly the collision statistic K_k of §10, one member of a complete weighted family.

Absolute continuity and the tail of $W_{q,\alpha_-(q)}$ remain conjectural for the coherent q-adic environment. The companion paper states and studies this as its own conjecture [8]; we restate it here for the role it plays in O7 below.

Conjecture 3.8 (Density and tail of the tilted Syracuse measure; [8])**.** *$W_q := W_{q,\alpha_-(q)}$ is absolutely continuous and non-degenerate, with $\Pr(W_q > x) = x^{-\alpha_+(q)/\alpha_-(q)+o(1)}$.*

Theorem 3.9 (Tail index in the i.i.d. model)**.** *For the additive martingale at parameter $\alpha_-(q)$ in the matching i.i.d. branching model, $\Pr(W_\infty > x) \sim C_q x^{-\alpha_+(q)/\alpha_-(q)}$ as $x \to \infty$, by the implicit renewal theorem for generalized multiplicative cascades [20].*

The companion paper's real-tree root batteries give related, independent empirical support, though for a formally separate object: at $q = 3$, two independent estimators (Hill, extreme-value theory); at $q = 5$, a sample of 10^5 roots shows a threshold-stability plateau at the predicted value and a Vuong test that no longer favors a lognormal alternative. That evidence targets the growth scale factor of the arithmetic tree itself, a companion conjecture in [8] expected to share Conjecture 3.8's exponent by the same implicit-renewal mechanism, without a proven equivalence between the two objects. A direct, exact test of W_q's own tail there (summing over the entire population of q-adic residues, no sampling involved) remains inconclusive. The companion paper gives the full battery of estimators, both routes, and the reasons neither is yet decisive for $q \geq 5$ [8].

Independent depth-first enumeration of the true reverse trees for $q = 5$ and $q = 7$ gives direct empirical support for Conjecture 3.5: counting slopes per decade of 3 sampled roots gave ≈ 0.62–0.66 for $q = 5$ (predicted 0.650919) and ≈ 0.36–0.39 for $q = 7$ (predicted 0.373501). A statistically calibrated measurement for $q = 5$ is in [9].

Remark 3.10 (Bibliographic calibration)*.* Neither the value $\alpha^*(5) = 0.650919$ nor the qualitative transition at $q \geq 5$ is new: Wirsching conjectured the transition heuristically in 1998 [4], and Kontorovich–Lagarias had already computed the same quantity (their $\eta_{5,BP}$, Theorem 8.10) via large-deviations methods for a branching random walk, a route different from, but numerically identical to, the pressure-equation derivation [11]. What is new here is the closed-form equation (4) itself, unifying the family for arbitrary q in a single algebraic identity, with its annealed derivation and independent numerical confirmation.

4 The endogeny barrier

4.1 Gauge freedom in the tree recursion

The recursion (2) is linear in G, and linearity has an immediate structural consequence that is easy to overlook: if W denotes the "arithmetic" solution (the one actually realized by, and measurable with respect to, the 3-adic digits of the tree) and Y is *any* bounded positive function invariant along the tree (in particular, any function of an auxiliary coordinate carried unchanged from parent to child), then $X := W \cdot Y$ solves *exactly the same recursion.*

Proposition 4.1 (Gauge freedom)**.** *Let $\mathbb{Z}_3^* \times \mathcal{Y}$ be the product space on which the 3-adic digit dynamics acts on the first factor exactly as on the reverse tree, while passing an auxiliary coordinate $y \in \mathcal{Y}$, independent of the digit dynamics, unchanged to every child. $W > 0$ is the recursion's arithmetic solution (nonnegative by (2)'s nonnegative weights), $\mathcal{F}_m$ the σ-algebra of the first m digits, and Z any $\mathcal{F}_\infty$-measurable statistic with $\log W$, Z square-integrable and $0 < \mathrm{Var}(Z) < \infty$ (in particular $Z = -\log \mu$). For non-constant $Y : \mathcal{Y} \to [c, C]$ with fixed constants $0 < c \leq C < \infty$, the function $X(r, y) := W(r) \cdot Y(y)$ satisfies:*

(a) the same pressure equation (4) *(identical weights, identical roots $\alpha = 1, \alpha^*$);*

(b) the same tail index as W, whenever $\Pr(W > x)$ *is regularly varying;*

(c) the same linear regression slope $b := \mathrm{Cov}(\log X, Z)/\mathrm{Var}(Z)$*;*

(d) but $\mathrm{Var}[\log X \mid \mathcal{F}_m] = \mathrm{Var}[\log W \mid \mathcal{F}_m] + \mathrm{Var}[\log Y]$ *for every m, so* $\mathrm{Var}[\log X \mid \mathcal{F}_m] \to \mathrm{Var}[\log Y] > 0$ *as $m \to \infty$, even though* $\mathrm{Var}[\log W \mid \mathcal{F}_m] \to 0$ *(W is $\mathcal{F}_\infty$-measurable): the conditional variance floor never collapses.*

Proof. (a) The pressure equation (4) identifies α from the branch weights of the recursion alone, which is homogeneous linear; Y is carried unchanged from parent to child and never enters a weight, so $X_v = \sum_i (\mathrm{weight}_i) X_{vi}$ holds whenever $W_v = \sum_i (\mathrm{weight}_i) W_{vi}$ does, leaving the weight structure, and hence the roots $\alpha = 1, \alpha^*$, untouched. (b) Since $0 < c \le Y \le C$ pointwise and $W \ge 0$, $cW \le X \le CW$, so $\Pr(W > x/c) \le \Pr(X > x) \le \Pr(W > x/C)$ for every x: whenever W's tail is regularly varying with index $-\alpha$, this sandwich alone forces the same index on X, with no independence hypothesis needed. Since Y is in fact independent of W by construction and bounded, hence $\mathbb{E}[Y^{\alpha+\varepsilon}] < \infty$ for every $\varepsilon > 0$, Breiman's lemma [17] sharpens this to the exact asymptotic $\Pr(X > x) \sim \mathbb{E}[Y^\alpha] \Pr(W > x)$: the tail *index* is preserved exactly, the tail *constant* generally is not. (c) $\log X = \log W + \log Y$ with Y independent of $\mathcal{F}_\infty \supseteq \sigma(Z)$ gives $\mathrm{Cov}(\log X, Z) = \mathrm{Cov}(\log W, Z) + \mathrm{Cov}(\log Y, Z) = \mathrm{Cov}(\log W, Z)$, so b is the same for X and W; $\mathrm{Var}(\log X) = \mathrm{Var}(\log W) + \mathrm{Var}(\log Y)$ adds pure noise around the same regression line. (d) Y is independent of $\mathcal{F}_m$ for every m (it never passes through the digit dynamics), so $\mathrm{Var}[\log X \mid \mathcal{F}_m] = \mathrm{Var}[\log W \mid \mathcal{F}_m] + \mathrm{Var}[\log Y]$ by independence of the two terms; W's own conditional variance vanishes in the limit since W is $\mathcal{F}_\infty$-measurable by definition, leaving exactly $\mathrm{Var}[\log Y] > 0$. ☐

No statistic computed from pressure, tail index, or regression slope against μ distinguishes W from $W \cdot Y$: this is items (a)-(c) above. Item (d) is the point of the construction: despite that, the conditional variance floor $\mathrm{Var}[\log Y] > 0$ persists at every resolution, so X is not endogenous. The technical name for this phenomenon is *endogeny* [12]: collapse of the conditional variance is equivalent to X being measurable with respect to the σ-algebra generated by the digits ("endogenous"); Proposition 4.1 exhibits a non-endogenous solution. The construction proves that the recursion *alone* does not force exactness. It does not assert that a non-trivial Y actually occurs for the arithmetic object built from a single integer v, where Y would have to be manufactured from v itself rather than injected as an independent auxiliary coordinate.

4.2 A specific mechanism ruled out

One natural candidate mechanism for a non-trivial gauge factor is a persistent 2-adic "memory" of the root surviving deep into the tree. This is false:

Lemma 4.2 (No 2-adic memory)**.** *Let w be a child of v at exponent a in the reverse tree of the classical ($q = 3$) map, so $3w = 2^a v - 1$. Then*

$$w \equiv -3^{-1} \pmod{2^a},$$

a universal constant depending only on a, independent of v. Consequently, a descendant reached after a path with exponent sum A has its lowest A bits determined entirely by the path*, with zero information about the root.*

Proof. From $3w = 2^a v - 1$ we get $3w \equiv -1 \pmod{2^a}$, i.e. $w \equiv -3^{-1} \pmod{2^a}$ since 3 is a unit mod 2^a. Induction along the path, composing the relation at each generation, gives the claim. ☐

We verified Lemma 4.2 numerically for $a = 1, \dots, 10$ against more than 500 values of v per exponent, confirming the predicted universal constant exactly in every case (e.g. $a = 5 \Rightarrow w \equiv 21 \pmod{32}$ always; $a = 10 \Rightarrow w \equiv 341 \pmod{1024}$ always). The reverse tree destroys all 2-adic information about the root at every generation; there is no persistent 2-adic coordinate available to correlate with the 3-adic residue, ruling out this specific mechanism for a gauge factor.

4.3 What forces collapse, and what does not

In the fully idealized probabilistic model (i.i.d. branching with independent subtree contents), exactness *is* forced, by two classical mechanisms:

(a) **Classification of smoothing-transform fixed points** [13, 14]: non-endogenous solutions of the associated smoothing-transform fixed-point equation generically have a strictly smaller tail index than the endogenous one (in the regime relevant here, tail index 1 with infinite mean), which is excluded by our own data: the pressure equation gives $\alpha^* = 2$ exactly, matching two independent empirical measurements (a Hill estimator and extreme-value theory on block maxima).

(b) **Choquet–Deny rigidity** for the Archimedean phase: bounded harmonic functions of a random walk on an abelian group are constant, which eliminates any Archimedean gauge coordinate under the full i.i.d. hypothesis.

Theorem 4.3 (The recursion does not force endogeny)**.** *The smoothing-transform model associated with the exact weight recursion of the reverse tree admits non-endogenous solutions (harmonic gauge factors, Proposition 4.1; the* 2*-adic obstruction is excluded by Lemma 4.2) with a strictly positive conditional-variance floor* $\sigma_Y := \mathrm{Var}[\log Y] > 0$ *at every resolution, statistically indistinguishable from the endogenous solution by pressure, tail index, or any marginal we have tested. Consequently the exact weight recursion together with the statistics of the* 3*-adic digits does not by itself force the residual gauge dispersion to vanish.*

Proof. Take Y non-constant, bounded away from 0 and ∞, invariant along the tree (a two-valued gauge on the auxiliary coordinate suffices). Proposition 4.1 gives $X := W \cdot Y$: it solves the exact same weight recursion as W and agrees with it in pressure equation, tail index, and regression slope against μ (items (a)-(c)), while $\mathrm{Var}[\log X \mid \mathcal{F}_m] \to \sigma_Y > 0$ as $m \to \infty$ (item (d)), so X is not measurable with respect to the full digit σ-algebra, hence not endogenous by the criterion above. Lemma 4.2 rules out a persistent 2-adic memory of the root as the mechanism realizing such a Y for the true arithmetic object, without excluding one manufactured from v by some other means. X is a non-endogenous solution, statistically indistinguishable from W by every statistic tested, proving the claim. □

Remark 4.4 (Barrier reading, and epistemic status)*.* Informally this delimits a class of arguments: we are aware of no argument using only (i) the exact weight recursion and (ii) statistics of the 3-adic digits that does not factor through this smoothing-transform model, and hence none that can distinguish the endogenous from the non-endogenous solution. We do *not* formalize “uses only (i) and (ii)” as a well-defined class of techniques (in contrast to relativization or natural-proofs barriers in complexity theory), so this is a heuristic delineation of known approaches rather than a formal impossibility theorem. The 2-adic exclusion (Lemma 4.2) and the gauge construction (Proposition 4.1) that Theorem 4.3 rests on are both fully proved; Lemma 4.2 is additionally checked numerically against direct computation, as recorded above. The classification claim above (that any such argument factors through the smoothing-transform model) can be stated unconditionally as "standard machinery under [hypotheses], detailed verification pending" rather than as independently re-derived against our exact setup. We never assert that the residual gauge dispersion σ_Y vanishes for the true arithmetic object; only that the recursion alone cannot force it to. The missing ingredient that closes the gap *in the idealized model*, and which remains to be transferred to the arithmetic setting, is cancellation of the centered fine-frequency covariance after independently averaging the two path indices, together with equidistribution of the Archimedean phase along paths. Literal near-independence of paired leaf digits is false by Theorem 8.2. The known rigorous analogue, in the forward direction, is the Fourier decay of Tao's Syracuse variables (Proposition 1.1); the reverse-tree/density version remains open and is *not* implied by it.

5 Empirical calibration of the residual coupling

Proposition 4.1 shows the gap is real in principle; it does not quantify how large the residual coupling between sibling subtrees actually is for the true arithmetic tree. We designed a three-arm control experiment to calibrate this directly.

5.1 Design

Arm 1. **Synthetic i.i.d. control**: a Monte Carlo model with the exact branch weights $q \cdot 2^{-a}$ but fully independent subtree contents, truncated by total population size (not depth) to mirror the true enumeration procedure.

Arm 2. **Synthetic coupled control**: the same model, but with sibling subtree *contents* coupled at adjustable intensity $\rho \in [0, 1]$ via whole-family replication addressed by a hash of (phase, replicate id, tag), constructed so that all single-subtree marginals remain exactly unchanged for any ρ.

Arm 3. **Real arithmetic tree**: exact enumeration of the true reverse tree, re-measured at matched truncation depth.

The residual variance excess, (Arm 3)−(Arm 1), compared against the calibration curve floor(ρ) built from Arm 2, converts a qualitative statement ("the gap exists") into a quantitative bound on the effective coupling ρ_{eff} that would be needed in the synthetic model to reproduce whatever residual is observed in the real tree.

5.2 Result

Across the full truncation grid $m = 8, \ldots, 29$, Arm 1 and Arm 3 showed nearly identical residual variances, with no sign of positive coupling. A final slope-and-bootstrap measurement at the deepest, best-calibrated truncation ($m = 20$, 4000 replicate roots, $\rho \in \{0, 0.05, 0.1, 0.2\}$) gives:

$$\rho_{\text{eff}} \lesssim 0.06 \quad (95\% \text{ CI}, \ m = 20). \tag{6}$$

The excess (real − synthetic) converges monotonically to zero with increasing truncation depth, the opposite sign from what a real positive arithmetic coupling would produce, reinforcing rather than merely failing to contradict the conclusion. This bound has a narrow scope: no coupling of the specific functional form explored by Arm 2 (whole-family replication of subtree *contents* at intensity ρ) was detected in the real tree beyond $\rho_{\text{eff}} \lesssim 0.06$ of that family's maximum intensity. This is not, by itself, a bound on every conceivable form of coupling; in particular it does not bound the exact, coarse, Δ-dependent pair correlation of Proposition 10.2 below, which lives on a disjoint axis (root residue rather than subtree content) already present identically, by construction, in Arm 1 itself; it is not a form of coupling this calibration could have detected or missed. See Remark 10.3 for the verified structural argument. Equation (6) converts the purely theoretical gap of Theorem 4.3 into a quantitative, falsifiable empirical bound on its size, for this family of couplings.

6 A finite-window comparison of the Kontorovich–Lagarias and Volkov predictions for $5x + 1$

Independently of the theoretical program above, a companion paper [9] reports an exact finite enumeration of the $5x+1$ reverse tree testing Conjecture 3.5 against a specific competing prediction.

Kontorovich–Lagarias [11] predict a counting exponent $\eta_{5,BP} \approx 0.650919$ for this tree (identical to $\alpha_-(5)$ above), while a competing stochastic model due to Volkov, discussed in the same reference, predicts $\eta^*_{5,BP} \approx 0.678$; the authors flag this as an open dispute, unresolved for lack of data able to distinguish the two, which differ by only $\Delta = 0.027$.

Empirical Result 6.1 (Finite-window comparison of two exponent predictions)**.** *The fixed-window estimator of the* $5x+1$ *reverse-tree counting exponent, computed by exact enumeration with Aitken-*Δ^2*-extrapolated truncation correction, is* 0.639 *(bootstrap interval* $[0.633, 0.645]$*). Since the sequence of local slopes has not stabilized within the tested windows, this alone does not exclude the Volkov prediction.*

The companion paper shows this estimator is itself biased, by more than the gap Δ in dispute, when run on a process of known exponent, so the raw reading above cannot settle the question. Building synthetic controls that share the arithmetic tree's branching and sibling structure (via an exact sibling congruence) but target a chosen exponent gives a calibrated comparison instead.

Empirical Result 6.2 (Calibrated separation from the Volkov exponent)**.** *At the truncation depth where calibrated controls read their own target exponent to within* 0.003*, the arithmetic tree reads* 0.64926 *(*95% *interval* $[0.64818, 0.65027]$*), separated from a control built to exponent* 0.678 *by seven band-widths, and matching controls built to exponent* 0.650919*. The separation widens as bias is calibrated away with depth; it never narrows. This measurement compares against the exponent value* 0.678 *alone, since Volkov's branching model itself is not implemented.*

7 Direct computational test of the Weak Covering Conjecture

7.1 Wirsching's conjecture

Wirsching [4] identifies the following combinatorial covering property as sufficient for uniform positive predecessor density (his Reduction Theorem V.3.10, building on the Weak Covering Conjecture V.3.9). Define

$$R_{j,k} := \left\{ \sum_{i=0}^{j} 2^{\alpha_i} 3^i \ : \ j+k \geq \alpha_0 > \alpha_1 > \cdots > \alpha_j \geq 0 \right\} \subset \mathbb{Z}_3^*, \qquad |R_{j,k}| = \binom{j+k+1}{j+1}.$$

Every element of $R_{j,k}$ is automatically a unit mod 3 (its top-order term is not divisible by 3, every other term is), so "covering $\mathbb{Z}_3^*$ mod 3^ℓ" means hitting all $2 \cdot 3^{\ell-1}$ invertible residue classes.

Conjecture 7.1 (Weak Covering Conjecture; Wirsching 1998, V.3.9)**.** *There is a sub-exponential function $K(\ell) > 0$ (i.e. $\lim_\ell K(\ell)e^{-\gamma\ell} = 0$ for every $\gamma > 0$) such that, for all j, ℓ: $|R_{j-1,j}| \geq K(\ell) \cdot 3^\ell \Rightarrow R_{j-1,j}$ covers $\mathbb{Z}_3^* \bmod 3^\ell$.*

For $j \geq \ell$, terms of index $i \geq \ell$ vanish mod 3^ℓ, giving the useful reduction $R_{j-1,j} \bmod 3^\ell = 2^{j-\ell} \cdot R_{\ell-1,j} \bmod 3^\ell$, which cuts the enumeration cost from $\binom{2j}{j}$ to $\binom{j+\ell}{\ell}$.

7.2 Implementation and validation

We implemented a cyclic-rotation bitset dynamic program: processing candidate exponents in decreasing order, we maintain, for each count c of terms chosen so far, the set of reachable partial sums mod 3^ℓ as a single Python integer bitset; choosing the exponent v at position c is a cyclic rotation by $2^v \cdot 3^c \bmod 3^\ell$, and population count of the final bitset gives the image size directly. The implementation matched an independently computed reference table exactly ($j^*(\ell)$ for $\ell = 1, \ldots, 7$: $1, 4, 6, 7, 9, 10, 11$) and a pointwise check ($\ell = 2, j = 2$: image $= \{1, 2, 5, 7\} \bmod 9$, missing $\{4, 8\}$).

7.3 Results

We computed $j^*(\ell)$, the smallest j for which $R_{j-1,j}$ covers, for $\ell = 1, \ldots, 20$ (cost grew by a factor ≈ 3.3 per step near the end; $\ell = 20$ alone took ≈ 27 minutes; further extension in pure Python was judged not worthwhile). Writing $e(\ell) := j^*(\ell) - \log_4 3 \cdot \ell$ for the excess over the entropy-matched threshold $\log_4 3 \approx 0.7925$:

ℓ	j^*	j^*/ℓ	$e(\ell)$
1	1	1.0000	0.208
5	9	1.8000	5.038
10	15	1.5000	7.075
15	20	1.3333	8.113
16	20	1.2500	7.320
17	21	1.2353	7.528
18	22	1.2222	7.735
19	23	1.2105	7.943
20	24	1.2000	8.150

(Full table for all 20 values is in the accompanying computational record.) $j^*(\ell)$ exists and is finite throughout the tested range. Across all 20 points, the local slope oscillates in 0.74–0.86, straddling the threshold $\log_4 3$ from both sides rather than converging cleanly; an isolated plateau at $\ell = 16$ crossing the sliding measurement window is what makes the first 17 points alone look like a monotonically decelerating slope approaching $\log_4 3$.

Model comparison on the tail $\ell = 10$–20 (constant vs. logarithmic vs. square-root vs. slow-linear growth of $e(\ell)$, by AIC) now disfavors pure stabilization ($\Delta\text{AIC} \approx 5$), consistent with a plateau-frequency test (only 1 plateau observed in 19 increments versus ≈ 3.9 expected under stabilization at rate $\log_4 3$; binomial $p \approx 0.072$, marginal but in the same direction). The four slow-growth models remain statistically indistinguishable from one another in this range ($\Delta\text{AIC} < 0.5$); they would only separate by ≥ 1 unit of j^* near $\ell \approx 40$, out of reach of this algorithmic approach.

Empirical Result 7.2 (Direct computational test of the Weak Covering Conjecture)**.** *$j^*(\ell)$ exists and is finite for all $\ell \leq 20$ (the first direct computation here of the exact object of Conjecture 7.1). The excess $e(\ell)$ qualitatively favors unbounded slow growth over stabilization (two* *independent statistics, ΔAIC and plateau frequency, in the same direction but not individually decisive), consistent with the weak form (Conjecture 7.1) and disfavoring the corresponding strong form with $K(\ell)$ bounded.*

8 Three precision regimes for Tao's bivariate extension

8.1 Why the naive bivariate argument fails

A natural strategy for closing the gap of Theorem 4.3 is to apply Proposition 1.1 twice, once per sibling leaf, and combine via some form of conditional independence given the common ancestor. A close reading of the mechanism of Tao's Section 7 proof (characters χ_ξ, i.i.d. Pascal-distributed valuation pairs, a phase evolving multiplicatively by $\log 9/\log 2$, a "black set" of well-separated triangles, a two-dimensional renewal process with monotonicity) shows that this mechanism operates entirely on $\mathrm{Syrac}(\mathbb{Z}/3^n\mathbb{Z})$ as constructed from *independent* geometric digits by definition (1); the link to actual integers is made elsewhere in [1] (via a total-variation approximation valid only for depth $n \lesssim \log N$).

For our problem, a single fixed integer v with two sibling leaves $w_1(v), w_2(v)$, the two leaves are *not* two i.i.d. samples; they are two different deterministic functions of the *same* variable v.

Proposition 8.1 (Perfect dependence at fixed depth)**.** *For a fixed pair of sibling paths of depth D, the existence congruence pins v to a single class mod 3^D: writing $v = v_0 + 3^D t$, the two leaves become affine functions of the same free variable t, with unit multipliers: $w_i = w_i(v_0) + 2^{A_i} t$. Consequently the pair $(w_1, w_2) \bmod 3^\ell$ lies on a line of size 3^ℓ inside $(\mathbb{Z}/3^\ell\mathbb{Z})^2$, for every $\ell \geq 1$: perfect deterministic dependence at every precision, with an entire line of resonant frequencies where the correlation has modulus exactly 1.*

Theorem 8.2 (Maximal coupling of fresh digits)**.** *Fix a pair of paths and condition their leaf values modulo 3^r. If the free arithmetic parameter is uniform over the 3^s compatible lifts modulo 3^{r+s}, let X_s, Y_s denote the next s base-3 digits of the two leaves. Then both marginals are uniform and*

$$d_{\mathrm{TV}}\big(\mathcal{L}(X_s, Y_s), \mathcal{L}(X_s) \otimes \mathcal{L}(Y_s)\big) = 1 - 3^{-s}, \qquad I(X_s; Y_s) = s \log 3.$$

Thus fresh digit blocks of a fixed leaf pair become maximally, rather than weakly, dependent as $s \to \infty$.

Proof. By Proposition 8.1, after the coarse class is fixed the two leaves are affine functions of one lift parameter, and both multipliers are units modulo 3^s. The map from the fresh block of the first leaf to that of the second is therefore a permutation of 3^s points. The joint law is uniform on its graph, while the product law is uniform on all 3^{2s} pairs. Direct summation gives total variation $1 - 3^{-s}$. Since one block determines the other, their mutual information equals the entropy of a uniform 3^s-point variable, namely $s \log 3$. $\square$

Both identities were checked numerically.

Consequently, "apply Proposition 1.1 twice, assuming conditional independence given the ancestor" is circular. Worse, it is false as stated, at any precision.

8.2 The correct, non-circular reformulation

The quantity that actually matters, decorrelation of *subtree aggregates* rather than leaf-to-leaf, is a second-moment statement, expanding into a sum over *pairs of paths*:

$$\mathbb{E}_v[Z_1 Z_2] \;\propto\; \#\{(a, a') : a_1 \in B_1,\ a'_1 \in B_2,\ \mathrm{Syrac}(a) \equiv \mathrm{Syrac}(a') \bmod 3^D\},$$

which by Fourier inversion gives Cov $\propto \sum_{\xi\neq 0} S_1(\xi)S_2(\xi)^*$, with S_i character sums conditioned on the first step. Here there is no circularity: the two path indices owe their independence to the tautology of expanding a square alone, no arithmetic hypothesis involved.

8.3 The three regimes

1. **Fixed pair, any ℓ**: perfect dependence (Proposition 8.1); no decay is possible, and decorrelation can only come from averaging over paths, never leaf-to-leaf.
2. $\ell = O(\log D)$: *provable*, with Tao's Section 7 machinery essentially unchanged. The budget $|\rho(\xi)| \leq C_A D^{-A}$ closes the sum over $3^\ell - 1$ frequencies provided $3^\ell \ll D^{2A}$, i.e. polynomial moduli in D: a reachable lemma, though its most natural formulation turns out to be false as originally sketched (§10).
3. $\ell \asymp c \cdot D$ (where the endogeny barrier, the fresh digits, and Wirsching's Weak Covering Conjecture all live): this requires exponential control of the linear-precision frequency tail after sublinear-conductor modes have been separated; Proposition 1.1 gives only super-polynomial decay, and Tao's Section 7 method structurally cannot give more, because the losses in the black-set argument are tied to Diophantine properties of $\log 3/\log 2$ via the slope $\log 9/\log 2$.

Remark 8.3 (Nearby open theories)*.* Regime 3 has features in common with correlations of digital functions in bases 2 and 3 [26], effective rigidity for $\times 2$, $\times 3$ actions, and Fourier decay for self-similar measures. We do not prove a reduction to or from any of these problems. The target used below is an exponential bound for the linear-precision part of the conditioned character sums $S_i(\xi)$, with sublinear-conductor modes treated through Theorem 9.11.

9 Triangulation: connections among three formulations of the obstruction

9.1 $\beta = 1$ (Tao 2020) and a weighted covering problem

Tao [3] defines $c_n := \inf_{b\in\mathbb{Z}/3^n\mathbb{Z},\, 3\nmid b} \Pr(\mathrm{Syrac}(\mathbb{Z}/3^n\mathbb{Z}) = b)$ and proposes the conjecture $\beta = 1$: $c_n = 3^{-n+o(n)}$ ("no residue class has probability much smaller than average"), proving conditionally that $\beta = 1$ implies Collatz density $\gg x^{1-o(1)}$ (almost total), and noting that the best known unconditional bound is Krasikov–Lagarias' $\gamma = 0.84$ [15].

Proposition 9.1 (Support identity and geometric cost)**.** *After a unit twist, Tao's Syracuse random variable and Wirsching's sets $R_{j,k}$ are the same algebraic object (sums of powers of 2 and 3 with strictly decreasing exponents). Let $B_\ell(z)$ be the least total geometric cost $a_1 + \cdots + a_\ell$ of a Syracuse tuple representing $z \pmod{3^\ell}$, and let $j^*(\ell)$ be the least j for which $R_{\ell-1,j}$ covers every admissible residue. Then*

$$\max_z B_\ell(z) = \ell + j^*(\ell).$$

More specifically, tuples of cost at most $\ell + j$ are in bijection, after multiplication by $2^{\ell+j}$, with $R_{\ell-1,j}$ modulo 3^ℓ.

Ordinary covering controls the cheapest representation of each residue, whereas c_ℓ sums the geometric weights of all its representations. At the WCC threshold $j^*(\ell) = \log_4(3)\ell + o(\ell)$, the cheapest representation alone gives

$$c_\ell \geq 2^{-\ell-j^*(\ell)} = 3^{-(1/2+\log_3 2)\ell+o(\ell)}.$$

This yields only $\beta \le 1/2 + \log_3 2 = 1.1309\ldots$, rather than $\beta = 1$ (identity and bound checked against brute force). The missing statement is a weighted covering estimate reaching the typical cost window: the total geometric multiplicity must be within a subexponential factor of its mean uniformly over residues. Our computation of $j^*(\ell)$ tests WCC, but does not by itself test this weighted assertion.

Theorem 9.2 (Large-deviation barrier at the WCC cost)**.** *Let $A_1, \ldots, A_\ell$ be independent with $\Pr(A_i = m) = 2^{-m}$ for $m \ge 1$, and put $B_\ell = \sum_i A_i$. For $1 < s < 2$,*

$$\Pr(B_\ell \le s\ell) = \exp\{-I(s)\ell + o(\ell)\}, \qquad I(s) = s\log 2 + (s-1)\log(s-1) - s\log s > 0. \tag{7}$$

Let $\mu_\ell^{(\le s)}$ be the subprobability Syracuse measure formed from tuples with $B_\ell \le s\ell$. Then, with the minimum over unit residues,

$$\min_a \mu_\ell^{(\le s)}(a) \le 3^{-\ell}\exp\{-I(s)\ell + o(\ell)\}.$$

At $s_* = 1 + \log_4 3$, the rate is $I(s_*) = 0.0120393866\ldots$, and the corresponding exponent relative to base 3 is

$$1 + \frac{I(s_*)}{\log 3} = 1.0109587219\ldots > 1.$$

Thus even perfect weighted equidistribution inside the critical WCC cost slice cannot establish $\beta = 1$.

Proof. The exact negative-binomial formula is

$$\Pr(B_\ell = m) = \binom{m-1}{\ell-1}2^{-m}, \qquad m \ge \ell.$$

For $m < 2\ell - 2$ these masses increase with m. Hence the logarithm of their sum through $\lfloor s\ell \rfloor$ differs from the logarithm of the final term by at most $O(\log \ell)$. Stirling's formula applied to that term gives (7). The total mass of $\mu_\ell^{(\le s)}$ is the probability in (7). Its minimum over the $2 \cdot 3^{\ell-1}$ unit residues is at most its average. The fixed positive factor between this average and $3^{-\ell}$ is absorbed by $o(\ell)$. Direct substitution of s_* gives the stated constants. □

The finite-level rate and bound are computed in the accompanying repository.

Theorem 9.2 sharpens the distinction between O2 and O3, two of the eight tracked points enumerated in the concluding §13, referenced by this (On) notation throughout the remainder of the paper. Multiplicity within the WCC budget can improve the single-representation exponent $1.1309\ldots$ only as far as the average truncated exponent $1.01096\ldots$. A route to $\beta = 1$ must reach costs $B_\ell = 2\ell - o(\ell)$, where the cost law has subexponential mass. This is the central microcanonical regime of Theorem 9.8.

Tao's 2020 approach also shows a second path exists alongside the endogeny/decorrelation route of §4: it replaces "near-independence between subtrees" with "deterministic separation of preimages (easy) plus L^∞ marginal control ($\beta = 1$, where all the difficulty lives)". This confirms that a route avoiding decorrelation altogether exists, and that it hits the *same* wall in marginal form.

A direct, unconditional bound on the worst-cylinder mass itself comes from an inter-level identity, independent of the microcanonical machinery above. Write $N_\ell(u) := 3^\ell \mu_\ell(u)$, so $c_\ell = \min_u N_\ell(u)/3^\ell$, and define $\beta_{\text{eff}}(\ell) := -\log_3(3^\ell c_\ell)/\ell + 1$, the exponent for which $3^\ell c_\ell = \ell^{-\Theta(1)} \cdot 3^{-\ell(\beta_{\text{eff}}(\ell)-1)}$ matches the observed decay; $\beta_{\text{eff}} \to 1$ is the target the weighted covering estimate needs. A one-term bound already gives $\beta_{\text{eff}} \le 2.523719$ (the first unit position in the defining window of §9 lies at index $j \le 1$, so $c_\ell \ge c_{\ell-1}/16$), and summing the geometric weight over every unit position in the least favorable phase improves this to $\beta_{\mathrm{eff}} \le 2.306270$ (weight $5/63$ instead of $1/16$). Both bounds use only a single level's worst-case value; the identity below uses an inter-level quantity instead.

For u a unit mod 3^ℓ, write $R_\ell(u) := N_\ell(u)/N_{\ell-1}(u \bmod 3^{\ell-1})$.

Theorem 9.3 (The inter-level cascade factor is nondecreasing)**.** $\min_u R_\ell(u)$ *is nondecreasing in* ℓ. *Consequently, for any fixed level* L *and every* $\ell \ge L$,

$$3^\ell c_\ell = \min_u N_\ell(u) \ge (\min_u R_L(u))^{\ell-L} \min_u N_L(u), \qquad \limsup_\ell \beta_{\mathrm{eff}}(\ell) \le 1 + \frac{\log(1/\min_u R_L(u))}{\log 3}.$$

Proof. Unrolling the memoryless recursion $\mu_\ell(y) = \frac{1}{2}\nu(2y) + \frac{1}{2}\mu_\ell(2y)$ along the 2-power orbit of y shows the valid exponents $t \ge 1$ with $2^t y \equiv 1 \pmod 3$ share a fixed parity $t_0(y) \in \{1,2\}$, and the step $t \to t+2$ acts on $z = 2^t y \bmod 3^\ell$ as $z \mapsto 4z$; writing $z = 1 + 3k$ this is the affine map $A(k) = 4k + 1 \bmod 3^{\ell-1}$, a single cycle of length $3^{\ell-1}$. This gives the exact identity $N_\ell(y) = 3 \cdot 2^{-t_0(y)} W_\ell(k_0(y))$, where $k_0(y) \in \mathbb{Z}/3^{\ell-1}\mathbb{Z}$ is y's position on this orbit and $W_\ell(k) := \sum_{j\ge 0} 4^{-j} N_{\ell-1}(A^j k)$.

The scale factors cancel in R_ℓ because t_0 and k_0 are *exactly* compatible between adjacent levels, and this is now proved, not merely checked. Since $t_0(y)$ depends only on $y \bmod 3$ and $y' := y \bmod 3^{\ell-1} \equiv y \pmod 3$, immediately $t_0(y') = t_0(y) =: t$. Write $z := 2^t y \bmod 3^\ell$ and $z' := 2^t y' \bmod 3^{\ell-1}$; since $y' \equiv y \pmod{3^{\ell-1}}$, also $z' = z \bmod 3^{\ell-1}$. As $z \equiv 1 \pmod 3$, write $z = 1 + 3k_0(y)$ with $k_0(y) \in [0, 3^{\ell-1})$; reducing $z - 1 = 3k_0(y)$ modulo $3^{\ell-1} = 3 \cdot 3^{\ell-2}$ gives $z' - 1 = (z \bmod 3^{\ell-1}) - 1 = 3\big(k_0(y) \bmod 3^{\ell-2}\big)$, hence $k_0(y') = (z'-1)/3 = k_0(y) \bmod 3^{\ell-2}$, the claimed compatibility.

Consequently $R_\ell(y) = W_\ell(k)/W_{\ell-1}(k \bmod 3^{\ell-2})$ with $k = k_0(y)$, and numerator and denominator run over the *same* index j with the same weights 4^{-j}, since A is given by an integer formula and so iterating it commutes with reduction mod $3^{\ell-2}$. Writing $k' := k \bmod 3^{\ell-2}$ and substituting $N_{\ell-1}(A^j k) = N_{\ell-2}(A^j k') R_{\ell-1}(A^j k)$ exhibits $R_\ell(y)$ as a weighted average of the values $R_{\ell-1}(A^j k)$, with weights $4^{-j} N_{\ell-2}(A^j k')/W_{\ell-1}(k') \ge 0$ summing to 1; a weighted average lies between the minimum and maximum of what it averages, which proves $\min_u R_\ell(u) \ge \min_u R_{\ell-1}(u)$. The displayed consequence follows by telescoping $N_\ell(u) = N_{\ell-1}(u \bmod 3^{\ell-1}) R_\ell(u)$ down to level L. Full derivation and the exact rational arithmetic certifying the bound below are in the accompanying repository. $\square$

At level $L = 10$, exact rational arithmetic certifies $\beta_{\mathrm{eff}} \le 1.882712$, improving the two single-level bounds above. The rate saturates: measured in floating point through $\ell = 16$, $\min R_\ell$ climbs toward ≈ 0.3803 against an observed ratio $\min N_\ell / \min N_{\ell-1}$ running at 0.93–0.97, so this route alone floors out near $\beta_{\mathrm{eff}} \approx 1.88$ and does not reach 1. Closing that gap is O2's remaining content: a pairwise anti-concentration inequality bounding the joint lower tail $\Pr(N_\ell(k) \le x \text{ and } N_\ell(A(k)) \le x)$ for consecutive units on the same orbit, uniformly down to $x \sim \exp(-c_0\ell)$, would already improve $\beta_{\mathrm{eff}} \le 1.88$ toward 1 for any fixed exponent in such a bound, via a union bound over the $\sim 3^{\ell-1}$ pairs combined with the exact recursive identity $W_\ell(k) = N_\ell(k) + \frac{1}{4}W_\ell(A(k))$ used above; it is a sufficient route to $\beta_{\mathrm{eff}} \to 1$, not shown here to be necessary.

A natural attempt at that pairwise inequality, a 3-adic Weyl-sum estimate for the mixed character sums attached to the two pair types of O2, fails for a structural reason. The relevant sums achieve the maximal square-root cancellation available at every primitive frequency, but the operator $T/|G|$ they define is, in the character basis, the matrix of composition with a bijection of the underlying group, hence unitary. Composing with a bijection is an L^2 isometry, so the route returns exactly the trivial Cauchy–Schwarz bound at every density, and cannot be nontrivial for the specific pair the union bound needs: no bound depending only on the two set sizes and spectral data of that bijection can close O2's remaining gap, however strong the individual character sum estimates are.

Proposition 9.4 (The pairwise inequality forces non-collapse). *If the pairwise anti-concentration inequality above holds, the q-adic density martingale at $\alpha = 1$, $q = 3$ (Theorem 3.6) does not converge to* 0 *almost surely under Haar measure.*

Proof. The parity class $t_0(y) \in \{1, 2\}$ of the proof of Theorem 9.3 depends only on $y \bmod 3$, splitting the unit group into two classes of Haar measure exactly 1/2 each, independent of ℓ. On the fixed class $t_0(y) = t_0$, the map $y \mapsto k_0(y)$ of that proof is a bijection onto $\mathbb{Z}/3^{\ell-1}\mathbb{Z}$ carrying the conditional Haar measure to the uniform measure, and $A(k) = 4k + 1$ is a further measure-preserving bijection of $\mathbb{Z}/3^{\ell-1}\mathbb{Z}$. Almost-sure convergence of $N_\ell(U)$, established over the whole unit group by Theorem 3.6, restricts to almost-sure convergence on this positive-probability class. If the limit were 0 a.s. there, $\Pr(N_\ell(k) \le x) \to 1$ for every fixed $x > 0$, k uniform on $\mathbb{Z}/3^{\ell-1}\mathbb{Z}$; since A preserves that uniform measure, a union bound on the complement gives $\Pr(N_\ell(k) \le x \text{ and } N_\ell(A(k)) \le x) \ge 1 - 2\Pr(N_\ell(k) > x) \to 1$ as well, contradicting any bound $Cx^\theta < 1$ (for some fixed exponent $\theta > 0$) on that joint probability once ℓ is large enough. □

The pairwise inequality is accordingly not obviously equivalent to $\beta_{\text{eff}} \to 1$: it rules out the one scenario, $N_\ell(U) \to 0$ a.s., under which the limiting measure at $q = 3$ would be Haar-singular, so it may be a strictly stronger statement than the Weak Covering Conjecture itself needs. A finite check gives no reason to expect that scenario here either: through $\ell = 16$, a power-law fit of $\Pr(N_\ell(U) \le 0.1)$ and of the increments of $\mathbb{E}[-\log N_\ell(U)]$ sits on the summable side of the exponent that would separate the two scenarios; thirteen levels do not separate a summable exponent from one merely close to the boundary, so this is corroborating evidence rather than a decisive test.

Theorem 9.3, the certified bound, the saturation measurement, the Weyl-sum obstruction, and Proposition 9.4 are all verified in the accompanying repository.

9.2 Wirsching's 2003 functional equation and the base-3 Fabius function

A different, more refined route toward the same goal is Wirsching [5]. This subsection restates, without proof, the results a companion paper [10] establishes in full. Its central combinatorial object, the "Elka" functions $e_\ell(k, a) := |E_{\ell,k}(a)|$, count paths of length $k + \ell$ in the Collatz graph ending at a, the same counting object underlying our entire $G(v)$/Syrac program, now viewed on the state space $\mathbb{X} = \mathbb{I}_3 \times \mathbb{Z}_3^\times$: the $\mathbb{Z}_3^\times$ factor carries the 3-adic variable (habitat of the Syracuse measure), while the $\mathbb{I}_3 \cong \{0, 1, 2\}^{\mathbb{N}}$ factor carries the normalized step-count $k/3^\ell$, converging to a density φ.

Proposition 9.5 (Base-3 Fabius function). *φ is the density of $X = \sum_{j\ge1} 2U_j \cdot 3^{-j}$, U_j i.i.d. uniform on $[0, 1]$: the base-3 analogue of the Fabius function. It is the unique $L^1([0, 1])$ fixed point of the averaging operator $W_3 f(x) := \frac{3}{2}\int_{3x-2}^{3x} f(t)\,dt$, C^∞, piecewise polynomial away from the standard Cantor set, with certified geometric convergence $\|f_n - \varphi\|_1 \le 2^{-n+1}\|f_1 - f_0\|_1$.*

Wirsching's chain reduces uniform positive predecessor density to five conditions ($\star$1)–($\star$5), stated here in the form the companion paper derives directly from the primary source. Let $g_\ell(k, a)$ denote Wirsching's *generator* at cost k and residue a, related to the Elka function by $e_\ell = p_\ell * g_\ell$ for a fixed partition count p_ℓ, and let $\overline{g}_\ell(k)$ be its average over units a. Fix $\delta > 0$ and let (k_ℓ) range over integer sequences with $|\ell - k_\ell| \le \delta\sqrt{\ell}$ (the central-limit window).

($\star$1) $\liminf_\ell e_\ell(k_\ell, a)/\overline{e}_\ell(k_\ell) \ge \mu$, uniformly over $a \in \mathbb{N}$ (Positive Density's target);
($\star$2) $\liminf_\ell g_\ell(k_\ell, a)/\overline{g}_\ell(k_\ell) \ge \mu_1$, uniformly over $a \in \mathbb{N}$;
($\star$3) $g_\ell(k_\ell, a) \ge \eta\,\overline{g}_\ell(k_\ell)$ for every $\ell \ge \ell_0$ and every unit $a \in \mathbb{Z}_3^\times$;
($\star$4) $\liminf_\ell (W_3^\ell \chi_1)(x_\ell)/(W_3^\ell \chi_0)(x_\ell) \ge \mu$, uniformly for $(x_\ell) \in \widetilde{A}_\delta$, with $\chi_0 = \mathbf{1}_{[0,2/3]}$, $\chi_1 = \mathbf{1}_{[1/3,1]}$;
($\star$5) an explicit asymptotic for φ tested in Empirical Result 9.7 below.

Two implications are theorems proved in [5]: ($\star$1) $\Rightarrow$ Positive Density, and ($\star$3) $\Rightarrow$ ($\star$2). The remaining three implications, ($\star$2) $\Rightarrow$ ($\star$1) (Conjecture 1), ($\star$4) $\Rightarrow$ ($\star$3) (Conjecture 2), and ($\star$5) itself (Conjecture 3, with ($\star$5) $\Rightarrow$ ($\star$4) proved in [5]), are conjectural. The companion paper reads the chain directly from the primary source and finds that ($\star$3) and ($\star$2) are in fact the same inequality, differing only in quantifier ($\ell \ge \ell_0$ versus $\liminf_\ell$), and that ($\star$4), Conjecture 2's actual target, is a statement purely about the one-dimensional operator W_3 carrying no information about the residue coordinate $a \in \mathbb{Z}_3^\times$ that ($\star$3) needs.

Theorem 9.6 (Wirsching's Conjecture 1)**.** *Condition ($\star$2) implies condition ($\star$1), with a possibly smaller central-limit window (a smaller δ).*

Conjecture 2 remains open; ($\star$3) is the most concrete sufficient condition above Conjecture 3, an asymptotic assertion on $\varphi(z_\ell)/\varphi_0(z_\ell)$ tested numerically below.

9.3 Certified numerical test of Wirsching's Conjecture 3

This subsection restates, without independent verification of its own, a companion paper's certified numerical test [10]. Using the exact self-similarity $X \overset{d}{=} (2U + X)/3$, the moments $M_i = \mathbb{E}[X^i]$ satisfy an exact rational recursion, giving a zero-truncation-error evaluation of φ at depth ℓ up to $\ell = 500$ (error certified $\le 10^{-8}$).

Empirical Result 9.7 (Numerical test of Wirsching's Conjecture 3)**.** *The decisive ratio $L_\ell := 3^{1-\ell}\varphi(3x_\ell^+)/\varphi(x_\ell^+)$, measured for $\ell = 200, \dots, 500$ in the CLT window ($u \in [-2, 2]$), converges to the predicted value $2/3$ for every u tested. At the central sequence $u = 0$, the deficit $\ell \cdot (2/3 - L_\ell)$ stabilizes near 0.580 (0.5807 at $\ell = 200$, 0.5800 at $\ell = 500$), a coefficient matching Berg–Krüppel's own asymptotic φ_0. This supports Conjecture 3 with certified numerical error on the tested range. Conjecture 2 remains open; Theorem 9.6 settles Conjecture 1 independently of this computation.*

9.4 Microcanonical decomposition and equivalence of ensembles

A microcanonical decomposition connects Wirsching's generators $g_\ell(k, a)$ to the Syracuse law μ_ℓ directly.

Theorem 9.8 (Microcanonical decomposition)**.** *Let $g_\ell(k, a)$ be Wirsching's generator and μ_ℓ the Syracuse law modulo 3^ℓ. There is an exact identity*

$$\mu_\ell(a) = \Big(\prod_{i=0}^{\ell-1} \tfrac{1}{2(1-2^{-c_i})}\Big) \sum_k 2^{-k} g_\ell(k, a), \qquad c_i = 2 \cdot 3^i. \tag{8}$$

If ($\star$3) holds (a uniform lower bound $g_\ell(k, a) \ge \eta\,\overline{g}_\ell(k)$ on the central-limit window), then $\mu_\ell(a) \ge C_{\delta,\eta} 3^{-\ell}$ uniformly: condition ($\star$3) implies a uniform lower bound on every cylinder of the Syracuse law, the estimate the weighted covering formulation of §9 needs.

Proposition 9.9 (Conditional local-limit bridge)**.** *The ratio $g_\ell(k,a)/\overline{g}_\ell(k)$ factors exactly as a canonical-mass factor $G_{\ell,a}(1/2)/\overline{G}_\ell(1/2)$ times a conditional-cost-law ratio $Q_{\ell,a}(k)/\overline{Q}_\ell(k)$. If both factors are uniformly bounded below on the central-limit window, ($\star 3$) holds. The companion paper shows the available local limit theorems for random expanding dynamics [6, 7] do not remove the growing-resolution obstruction this bridge is subject to.*

The companion paper also proves a sharp trichotomy on how much of ($\star 3$) local-limit methods can reach: exact equivalence of ensembles at fixed precision (Theorem 9.10) and at sublinear precision (Theorem 9.11), against a matching nonequivalence at linear precision (Theorem 9.12), and a full-precision likelihood bound in the wrong direction for ($\star 3$) (Theorem 9.13).

Theorem 9.10 (Equivalence of ensembles at fixed precision)**.** *Fix $r \geq 1$. Uniformly on the central-limit window, the projection of $p_{\ell,k}(a) := g_\ell(k,a)/\sum_b g_\ell(k,b)$ modulo 3^r converges in total variation to μ_r as $\ell \to \infty$.*

Theorem 9.11 (Equivalence of ensembles at sublinear precision)**.** *For $r = r_\ell = o(\ell)$, the same projection has total-variation distance $o(1)$ from μ_r, uniformly on the central-limit window.*

Theorem 9.12 (Linear-block nonequivalence)**.** *If $r/\ell \to \rho \in (0,1)$, the terminal block of r folded costs, conditioned on the central-limit event, has total-variation distance from its unconditioned law converging to a strictly positive limit (two Gaussians of different variance). This shows $r = o(\ell)$ is sharp for the latent cost vectors; it does not by itself show nonequivalence after projection modulo 3^r, since the residue map can discard the block sum.*

Theorem 9.13 (Full-precision likelihood domination)**.** *For every $1 \leq r \leq \ell$ and every set E, the projected microcanonical law satisfies $p^{(r)}_{\ell,k}(E) \leq \Pr(K_\ell = k)^{-1}\mu_r(E)$, giving $D_{\mathrm{KL}}(p^{(r)}_{\ell,k}\|\mu_r) \leq \frac{1}{2}\log\ell + O_\delta(1)$ uniformly on the central-limit window: the relative entropy per digit vanishes, ruling out an exponential likelihood separation, but in the direction opposite to what ($\star 3$) needs (a lower bound on every residue, rather than an upper one).*

Empirical Result 9.14 (Fixed-precision projection test)**.** *Exact fixed-cost tables projected onto 3^r, $r = 1,2,3$, compared in total variation with μ_r: distances 0.02–0.06 at $\ell = 12$, central cost, decreasing over the computed levels; full precision at $\ell = 13$ gives 0.25–0.26, not yet determining an asymptotic limit.*

Empirical Result 9.15 (Finite microcanonical coverage)**.** *Exact multiplicities $g_\ell(k,a)$ through $\ell = 12$ (Boolean support through $\ell = 16$): coverage fraction of unit residues at the central cost rises from 0.81 ($\ell = 12$) toward 0.9999997 ($\ell = 16$, upper window), first fully covering cost $\ell + 5$ for $10 \leq \ell \leq 16$. The minimum multiplicity is still zero throughout; these values neither prove nor refute the uniform asymptotic lower bound ($\star 3$) needs.*

Empirical Result 9.16 (Fourier budget for fixed-cost multiplicities)**.** *Additive Fourier inversion bounds $\min_a p_{\ell,k}(a)$ below by 1 minus a spectral sum that grows from 12.1 ($\ell = 4$) to 293.3 ($\ell = 12$) at central cost, dominated by primitive frequencies; the resulting lower bounds are negative throughout. This shows termwise absolute summation of the finite Fourier series fails to establish uniform positivity on this range; whether positivity itself fails remains open.*

Together, Empirical Results 7.2 and 9.7 and Proposition 9.1 connect integer covering, the weighted Syracuse distribution, and a real functional equation. The support identity is exact; the required microcanonical multiplicity estimate and Wirsching's Conjecture 2 remain unresolved. Full proofs, the complete equivalence-of-ensembles program, and a new finding on central-cost support zeros are in the companion paper [10].

10 Two candidate lemmas, executed to completion

Sections 8–9 identify two natural next steps: a "regime-2 lemma" (decorrelation of subtree aggregates at logarithmic precision) and a "Z-number reduction lemma" (formalizing the analogy with Littlewood–Offord/Bohr-set methods from [2]). The first yields an exact obstruction and a finite test. The second retains an open frequency-stability step.

10.1 The regime-2 lemma: structure theorem and a finite test of its natural sufficient condition

The originally sketched budget, $|\rho(\xi)| \le C_A D^{-A}$ *uniformly in* ξ, contradicts Proposition 1.2 (an exact identity rather than a lossy estimate): for any frequency ξ of primitive level ℓ' (i.e. $\xi = 3^{\ell-\ell'}\xi_0$, $3 \nmid \xi_0$) and any depth $D \ge \ell'$, $\mathbb{E}[e(\xi F_D/3^\ell)] = \varphi_{\ell'}(\xi_0)$ is *independent of* D: depth beyond ℓ' is a dummy parameter, because 3-adic ultrametricity truncates the relevant sum to the coarsest ℓ coordinates (Proposition 1.2). A closed example at $\ell' = 1$: $\mathrm{Syrac}_1 = 2^{-a} \bmod 3 \in \{1, 2\}$ with probabilities $1/3, 2/3$; $|\varphi(1)| = 1/\sqrt{3} \approx 0.577$ for *every* D, with no decay whatsoever. The correct per-frequency budget is $C_A \ell'^{-A}$ (Proposition 1.1, which requires $3 \nmid \xi$), in place of $C_A D^{-A}$; even with this correction, the total ℓ^∞ budget summed over frequencies, $\sum_{\ell' \le \ell} 3^{\ell'}(C_A \ell'^{-A})^2 \asymp 3^\ell \ell^{-2A}$, never tends to 0: closing by frequency counting would require square-root cancellation, $\ell^{-\ell'/2-\varepsilon}$ decay, Regime 3 resurfacing inside what should have been the easy regime.

A structure theorem survives the failed budget. With $w_2 = 2^\Delta w_1 + c_\Delta$, $c_\Delta = (2^\Delta - 1)/3$ (a unit multiplier mod 3^ℓ) relating admissible sibling leaves at exponent gap Δ:

Lemma 10.1 (Covariance spectral identity)**.** *For subtree-aggregate observables $Z_i := N^{(m)}_{\eta_i}(w_i(v))$ (weighted populations truncated to precision m), with v uniform on admissible classes mod 3^{m+1},*

$$\mathrm{Cov}(Z_1, Z_2) = 3^{-2m} \sum_{\xi \not\equiv 0 \bmod 3^m} e(-\xi c_\Delta/3^m)\, S_1(-2^\Delta \xi)\, S_2(\xi),$$

by character orthogonality and the affine sibling relation. The two summation indices are independent by the tautology of expanding the square (not an arithmetic hypothesis), but the dependence through the common v is not thereby cancelled: it survives entirely as the diagonal frequency pairing $\xi_1 = -2^\Delta \xi_2$.

Proposition 10.2 (Exact coarse endogeny)**.** *For complete populations ($\eta_i \equiv 1$) and precision $\ell = 1$:* $\mathrm{Corr}(Z_1, Z_2) = +1$ *if* $\Delta \equiv 0 \pmod 6$*, and* $= -1/2$ *otherwise, independent of depth D and of any truncation.*

Proposition 10.2 converts the endogeny barrier of Theorem 4.3 from a qualitative argument into a closed form: aggregating leaves does *not* decorrelate them; aggregation projects onto the coarse σ-algebra generated by $w_i \bmod 3^\ell$, where the affine sibling relation of the fixed-pair regime (§8, Regime 1) reappears intact rather than vanishing. This is a positive result with the opposite sign from what was hoped for: what is provable is the persistence of coarse correlation; disappearance remains unproved.

Decomposing the sum of Lemma 10.1 by primitive level and applying Cauchy–Schwarz plus Parseval gives $\mathrm{Cov} = C_{\mathrm{exact}}(\ell_0; \Delta) + \mathrm{Err}$, with C_{exact} exact, finite, and D-independent (Proposition 10.2), and

$$\frac{|\mathrm{Err}|}{\mathbb{E}[Z_1]\mathbb{E}[Z_2]} \;\le\; K_m - K_{\ell_0}, \qquad K_\ell := 3^\ell \sum_y \mu_\ell(y)^2,$$

the normalized *collision mass* of the Syracuse measure μ, unrelated to Wirsching's covering function $K(\ell)$ of Conjecture 7.1: the subscript notation K_ℓ is reserved for the collision mass throughout the rest of this paper. A conditional lemma would follow if $K_\infty := \lim_\ell K_\ell < \infty$ (equivalently, the density $f = d\mu/d(\text{Haar})$ lies in $L^2(\mathbb{Z}_3)$): the fine covariance, net of the exact coarse component, would then be $\leq K_\infty - K_{\ell_0} \to 0$, uniformly in m, D, Δ.

Remark 10.3 (Consistency with the calibration of §5). Proposition 10.2 and the null result $\rho_{\text{eff}} \lesssim 0.06$ of §5 measure different objects. In the synthetic model underlying §5's Arm 1 and Arm 2, a child's type (its residue mod 3, the $\ell = 1$ object Z_i of Proposition 10.2) is a deterministic function of the root's phase and the branch index, reproducing the true cyclic admissibility structure and independent of the coupling intensity ρ that Arm 2 varies. Checked directly in the simulator's code (2000 seeds, $\rho \in \{0, 0.3, 0.7, 1\}$, zero divergence in the resulting type sequence), this deterministic type structure reproduces Proposition 10.2's exact correlation ($+1$ for $\Delta \equiv 0 \pmod 6$, $-1/2$ otherwise) to four decimal places. The coarse correlation is already present in Arm 1 ($\rho = 0$) exactly as in the real tree; ρ governs a disjoint axis, whether sibling subtrees' *content* is shared, distinct from whether their root residues correlate. The bound $\rho_{\text{eff}} \lesssim 0.06$ could not have detected Proposition 10.2's correlation in the first place.

A second, quantitative reason points the same way. Proposition 10.2's covariance $C_{\text{exact}}(\ell_0; \Delta)$ is finite and D-independent, while the aggregate means $\mathbb{E}[Z_i]$ grow exponentially with truncation depth, so its normalized contribution $C_{\text{exact}}/(\mathbb{E}[Z_1]\mathbb{E}[Z_2]) \to 0$ as depth increases: consistent with the excess of §5 decaying monotonically to zero. The finite-level L^2 computation below (Empirical Result 10.4) tests an upper-bound route; §5 measures the fine covariance itself directly and finds it small. Weighting branch pairs by their natural second-moment mass $2^{-a_i} \cdot 2^{-a_j}$, $\Delta = 2k$ has probability $3 \cdot 4^{-k}$, so $\Pr[\Delta \equiv 0 \pmod 6] = 1/21$ against $20/21$, giving mean coarse correlation $\mathbb{E}[\text{Corr}] = \frac{1}{21}(+1) + \frac{20}{21}(-\frac{1}{2}) = -\frac{3}{7}$, the same sign as the residual excess of §5. Theorem 4.3 and Proposition 10.2 establish that the recursion cannot force sibling decorrelation; the actual aggregate residual, on the orthogonal axis of subtree content, is nonetheless empirically small.

Empirical Result 10.4 (Finite-level growth of the L^2 statistic)**.** *K_ℓ was computed exactly via the recursion*

$$\mu_n(y) = \tfrac{1}{2}\,\nu(2y \bmod 3^n) + \tfrac{1}{2}\,\mu_n(2y \bmod 3^n), \tag{9}$$

where ν is the law of $1 + 3F_{n-1}$ (derived below), solved as a truncated geometric series along the cyclic orbit of multiplication by 2 mod 3^n (a single orbit, since 2 is a primitive root mod 3^n for every n). For $\ell = 1, \ldots, 17$, K_ℓ grows cleanly and consistently linearly, *with increments converging monotonically to ≈ 0.47, with no sign of saturation ($K_1 = 5/3$ exactly, matching a hand computation from* $\Pr(\text{Syrac}_1 = 1) = 1/3$, $\Pr(\text{Syrac}_1 = 2) = 2/3$*). These finite data are evidence against boundedness, but do not prove $K_\infty = \infty$.*

Derivation of the recursion (9)*.* By Proposition 1.2, $F_n(a) = 2^{-a_1}\big(1 + 3F_{n-1}(a_2, \ldots, a_n)\big) \bmod 3^n$. Write $X := 1 + 3F_{n-1}(a_2, \ldots, a_n)$, with law ν, independent of a_1. Conditioning on $a_1 = 1$ (probability $1/2$) gives $F_n = 2^{-1}X$; conditioning on $a_1 \geq 2$ (probability $1/2$), the memorylessness of the geometric distribution gives $a_1 - 1 \sim \text{Geom}(2)$ again, so $F_n = 2^{-1}\big(2^{-(a_1-1)}X\big)$, and the bracketed quantity has exactly the law of F_n itself (an independent copy). This gives the self-referential fixed point $\mu_n(y) = \frac{1}{2}\nu(2y) + \frac{1}{2}\mu_n(2y)$, all arithmetic mod 3^n. ☐

The $\ell = 1$ base case alone would not exercise the recursive machinery, so we validated the recursion itself against direct Monte Carlo sampling of Syrac from its primary definition (1) at $\ell = 3, 4$ (3×10^6 samples each), bin-by-bin: maximum discrepancy 0.000311 ($\ell = 3$) and 0.000213 ($\ell = 4$), both within the expected Monte Carlo noise ($\approx 1/\sqrt{N} \approx 0.000577$).

Empirical Result 10.5 (Finite-level L^p collision spectrum)**.** *We evaluated* (5) *for the Syracuse law at* $p \in \{1.1, 1.25, 1.5, 1.75, 1.9, 2, 2.1, 2.5\}$ *and* $k \leq 14$*, using the same recursion and checks as Empirical Result 10.4. At* $k = 14$*, the values for* $p = 1.1, 1.5, 1.9, 2, 2.1, 2.5$ *were respectively* 1.129*,* 2.186*,* 5.771*,* 7.743*,* 10.603*, and* 45.211*. Their last consecutive ratios were* 1.0017*,* 1.0161*,* 1.0512*,* 1.0647*,* 1.0804*, and* 1.1681*. Growth slows markedly below* $p = 2$ *and increases above it on this range. The data do not establish boundedness or divergence for any* p*.*

Both empirical results above, and the Monte Carlo validation of the recursion, are in the accompanying repository.

Remark 10.6 (Connection to a conjectural tail exponent). If the limiting Syracuse density exists and satisfies $\Pr(f(U) > x) \sim Cx^{-2}$, then its pth moment is finite for $p < 2$ and infinite for $p \geq 2$. The finite-level pattern in Empirical Result 10.5 is compatible with this threshold. It does not prove the existence of the density, the tail asymptotic, or the moment behavior at infinite level.

Remark 10.7 (Flatness questions for the Syracuse measure). Tao's oscillation estimate, boundedness of K_∞, and the L^∞-type conjecture $\beta = 1$ are distinct statements. The finite computation above does not settle the L^2 statement or either of the other two.

The regime-2 lemma is thus not an easier stepping stone toward Regime 3: it is a sibling of it, sharing the same missing ingredient, now with finite evidence against the natural sufficient condition.

10.2 The Z-number reduction: a conditional dichotomy and an exact negative result

The second candidate lemma formalizes "failure of the Weak Covering Conjecture at scale ℓ with margin c implies a Bohr-type concentration configuration", via the Littlewood–Offord/Riesz-product technique of Tao's 2011 blog post [2], a recurring source of miscitation as a paper. That post proves conditional separation results for $2^a - 3^k$ via Halász's method [29] and inverse Littlewood–Offord theory [30, 31], but acknowledges that Baker's theorem already gives an unconditionally stronger separation, so treats its own propositions as methodological exercises. Mahler's classical "Z-number" problem [32] concerns $\xi(3/2)^n$, a structurally adjacent but different object from ours ($\times 2 \bmod 3^s$ at finite depth, versus $\times 3/2 \bmod 1$ on an infinite orbit). Ours anchors instead in the Erdős/Lagarias family, including the general fractional-parts framework of Flatto–Lagarias–Pollington [34, 35, 33].

Definition 10.8 (Failure with margin c)**.** For $c > 0$, the Weak Covering Conjecture *fails at scale* ℓ *with margin* c if, for $j = \lceil (1 + c) \log_4 3 \cdot \ell \rceil$, some $b \in (\mathbb{Z}/3^\ell\mathbb{Z})^\times$ lies outside the image of $R_{j-1,j}$. Set $\gamma_c := (j + \ell)/\ell = 1 + (1 + c) \log_4 3$.

Definition 10.9 (Diagonal configuration)**.** A configuration of depth s at slope γ, tolerance δ, and exception rate η is $\xi \in \mathbb{Z}$, $3 \nmid \xi$, such that in $\geq (1 - \eta)s$ of the scales $t \in \{1, \dots, s\}$ there is $a \in [\gamma t - t^{2/3}, \gamma t + t^{2/3}]$ with $\|\xi \cdot 2^a/3^t\| < \delta$.

Lemma 10.10 (Pre-wrap triviality)**.** *For* $\xi = 1$*,* $\|2^a/3^s\| < \delta$ *holds trivially for* $a \leq \log_2 3 \cdot s + \log_2 \delta$ *(the representative has not yet wrapped around). Since* $\gamma_c \geq 1.7925 > \log_2 3 \approx 1.585$ *for every* $c \geq 0$*, the forced configuration lives entirely in the non-trivial post-wrap regime, by structural slack coming from the* 4^j *vs.* 3^ℓ *entropy accounting.*

Proposition 10.11 (Coarse modes and the sharp primitive hole barrier)**.** *Let $N = 3^\ell$, let $U = (\mathbb{Z}/N\mathbb{Z})^\times$, and suppose μ is any probability measure supported on U. Then*

$$|\widehat{\mu}(N/3)| \geq \frac{1}{2}.$$

For $\ell \geq 2$, fix $a \in U$ and let u_a be uniform on $U \setminus \{a\}$. This measure has a support hole, but for every primitive additive frequency $3 \nmid \xi$,

$$|\widehat{u}_a(\xi)| = \frac{1}{2 \cdot 3^{\ell-1} - 1}.$$

There are also probability measures with support holes for which every primitive coefficient is zero.

Proof. At frequency $N/3$, the coefficient is $pe(1/3)+(1-p)e(2/3)$, where p is the mass in the residue class $1 \pmod 3$. Its real part is $-1/2$, proving the first bound. For primitive ξ, the Ramanujan sum over all units modulo 3^ℓ is zero when $\ell \geq 2$. Removing a leaves

$$\widehat{u}_a(\xi) = -\frac{e(\xi a/N)}{|U| - 1},$$

which proves the second assertion. For the last assertion, take any probability measure on the units modulo $3^{\ell-1}$ whose support is not full and lift it uniformly to each of the three residues above every point. For $3 \nmid \xi$, summing the character over a fibre contributes $1 + e(\xi/3) + e(2\xi/3) = 0$. □

Proposition 10.12 (Primitive energy is fresh-fibre imbalance)**.** *Let μ be a probability measure on $\mathbb{Z}/3^\ell\mathbb{Z}$. For $b \bmod 3^{\ell-1}$ put*

$$x_t(b) = \mu(b + t3^{\ell-1}), \qquad t = 0, 1, 2.$$

Then

$$\sum_{3\nmid\xi} |\widehat{\mu}(\xi)|^2 = 3^{\ell-1} \sum_b \big((x_0 - x_1)^2 + (x_1 - x_2)^2 + (x_2 - x_0)^2\big). \tag{10}$$

Thus all primitive coefficients vanish if and only if μ is the uniform lift of its marginal modulo $3^{\ell-1}$. If $\mu(x) = n_x/T$ for nonnegative integer counts and at least one fibre is not uniform, then

$$\max_{3\nmid\xi} |\widehat{\mu}(\xi)| \geq \frac{1}{T}.$$

For $\ell \geq 2$, this bound is attained by the uniform law on all units except one.

Proof. Parseval at level ℓ gives

$$\sum_\xi |\widehat{\mu}(\xi)|^2 = 3^\ell \sum_x \mu(x)^2.$$

The nonprimitive coefficient at $\xi = 3\eta$ is the coefficient at η of the marginal $m(b) = x_0(b) + x_1(b) + x_2(b)$. Subtracting Parseval for that marginal gives

$$3^{\ell-1} \sum_b \left(3 \sum_{t=0}^{2} x_t(b)^2 - m(b)^2\right),$$

which is the right side of (10). If an integer fibre is nonuniform, the sum of its three squared pairwise differences is at least two. There are $2 \cdot 3^{\ell-1}$ primitive frequencies, so their maximum squared modulus is at least T^{-2}. The final equality case follows from Proposition 10.11. □

The identity also shows that projective compatibility does not repair the support-hole implication. Start with a non-full law at some coarse level and lift it uniformly through every subsequent three-point fibre. The resulting laws are projectively consistent and non-full, but every new primitive spectrum is zero. Primitive concentration measures imbalance of the newest digit, rather than inherited support failure.

Theorem 10.13 (Multiscale Parseval decomposition)**.** *Let $(\mu_r)_{r\geq 0}$ be compatible probability laws modulo 3^r, with μ_0 the point mass on the trivial group. Define*

$$K_r = 3^r \sum_x \mu_r(x)^2, \qquad E_r = \sum_{3\nmid\xi} |\widehat{\mu}_r(\xi)|^2.$$

Then

$$K_r - K_{r-1} = E_r, \qquad K_\ell = 1 + \sum_{r=1}^{\ell} E_r. \tag{11}$$

If the laws are the projections of a probability measure μ on $\mathbb{Z}_3$, then μ has an L^2 density with respect to Haar measure if and only if

$$\sum_{r\geq 1} E_r < \infty.$$

Proof. Parseval identifies K_r with the sum of $|\widehat{\mu}_r(\xi)|^2$ over all characters modulo 3^r. The characters with $3 \mid \xi$ are the pullbacks of the characters modulo 3^{r-1}. Compatibility identifies their coefficients with those of μ_{r-1}, so their total contribution is K_{r-1}. The remaining characters are primitive and contribute E_r. This proves (11) by subtraction and iteration.

The level-r density relative to Haar measure is the conditional density martingale of μ, and its squared L^2 norm is K_r. Boundedness of these norms is equivalent to L^2 martingale convergence. The limit is an L^2 density for μ. Conversely, if μ has an L^2 density, the level densities are its conditional expectations and have bounded L^2 norms. Equation (11) completes the equivalence. □

Theorem 10.13 identifies the exact overlap between O5 and the L^2 branch of O7. Proposition 10.12 describes the increment created by the newest digit, while K_ℓ is the accumulated energy through level ℓ. The identity neither selects the largest primitive coefficient at one level nor aligns frequencies across levels, so it does not prove Conjecture 10.18.

For the q-adic Syracuse laws at the geometric weight $\alpha = 1$, the one-level collision admits a second diagonalization. It uses Fourier analysis after ordering the preceding law by a discrete logarithm, rather than additive Fourier analysis on the residue group.

Theorem 10.14 (q-adic logarithmic collision renormalization)**.** *Let q be an odd prime, put $d = \operatorname{ord}_q(2)$, and suppose that $v_q(2^d - 1) = 1$. Let $\mu_{q,\ell}$ be the Syracuse law modulo q^ℓ with digit probabilities $\Pr(a) = 2^{-a}$, and define*

$$K_{q,\ell} = q^\ell \sum_x \mu_{q,\ell}(x)^2.$$

For $M = q^{\ell-1}$, put

$$a_j = \mu_{q,\ell-1}\left(\frac{2^{dj}-1}{q} \bmod M\right), \qquad A_m = \sum_{j=0}^{M-1} a_j e^{-2\pi i m j/M}.$$

Then, for every $\ell \geq 1$,

$$K_{q,\ell} = \sum_{m=0}^{M-1} \frac{q(4^d-1)|A_m|^2}{3(4^d+1-2^{d+1}\cos(2\pi m/M))}. \tag{12}$$

In particular, with

$$c_{q,d} = \frac{q(2^d-1)}{3(2^d+1)}, \qquad C_{q,d} = \frac{q(2^d+1)}{3(2^d-1)},$$

one has

$$c_{q,d}K_{q,\ell-1} \le K_{q,\ell} \le C_{q,d}K_{q,\ell-1}. \tag{13}$$

For $q \ge 5$, $c_{q,d} \ge 35/27 > 1$. Hence $K_{q,\ell} \ge c_{q,d}^{\ell}$, and the projective Syracuse law has no L^2 density on $\mathbb{Z}_q$.

Proof. The lifting assumption gives $\operatorname{ord}_{q^\ell}(2) = dM$. Write $y_k = 2^k \bmod q^\ell$, $x_k = \mu_{q,\ell}(y_k)$, and let n_k be the mass at y_k of the law of $1 + qF_{\ell-1}$. The Syracuse recursion gives

$$x_k = \frac{1}{2}n_{k+1} + \frac{1}{2}x_{k+1},$$

with indices modulo dM. Also, $n_{dj} = a_j$, and $n_k = 0$ when $d \nmid k$. The map $j \mapsto (2^{dj}-1)/q \bmod M$ is a permutation: equality of two outputs modulo M is equivalent to equality of the corresponding powers modulo q^ℓ, where 2^d has order M.

Set $u_j = x_{dj}$. A block of d recursion steps gives

$$u_j = 2^{-d}(a_{j+1} + u_{j+1}), \qquad x_{dj+r} = 2^r u_j \quad (0 \le r < d).$$

Hence $\sum_k x_k^2 = (4^d-1)\sum_j u_j^2/3$. At frequency $z = e^{2\pi i m/M}$, the circular filter from a to u has multiplier $z/(2^d - z)$, whose squared modulus is $(4^d + 1 - 2^{d+1}\cos(2\pi m/M))^{-1}$. Parseval and $q^\ell = qM$ prove (12). Another application of Parseval gives $K_{q,\ell-1} = \sum_m |A_m|^2$. The minimum and maximum of the multiplier occur at cosine -1 and 1, respectively, which proves (13). If $q \ge 5$, then $d \ge 3$, so $c_{q,d} \ge (5/3)(7/9) = 35/27$. $\square$

For $q = 3$, one has $d = 2$, and (12) reduces to

$$K_\ell = \sum_m \frac{15}{17 - 8\cos(2\pi m/M)}|A_m|^2.$$

Together with Theorem 10.13, this gives

$$E_\ell = \sum_m \left(\frac{15}{17 - 8\cos(2\pi m/M)} - 1\right)|A_m|^2.$$

The signed multiplier is positive exactly when $\cos(2\pi m/M) > 1/4$. Thus the unresolved $q = 3$ part of O7 is a balance between low and high frequencies of the preceding law in logarithmic order. The uniform lower bound is only $3/5$, so the exponential argument for $q \ge 5$ does not apply.

Proposition 10.11, Proposition 10.12, Theorem 10.13, and Theorem 10.14 above are all verified, including the general-q case for $q = 7, 11, 13, 17, 19$.

Theorem 10.14 needs q prime and the maximal lifting hypothesis $v_q(2^d - 1) = 1$. Conditioning on the first exponent instead of diagonalizing removes both restrictions.

Theorem 10.15 (Diagonal-term collision rate, unconditional)**.** *For every odd $q \ge 3$ (prime or composite), with $Y = 1 + qF_{\ell-1}$ and $H_{\ell-1}(s) := q^{\ell-1}\Pr[Y' = 2^s Y \bmod q^\ell]$,*

$$K_{q,\ell} = \frac{q}{3}\sum_{s\in\mathbb{Z}} 2^{-|s|}H_{\ell-1}(s), \qquad H_{\ell-1}(0) = K_{q,\ell-1},$$

with every term nonnegative. Dropping every $s \ne 0$ gives, for every $\ell \ge 1$,

$$K_{q,\ell} \ge \frac{q}{3}K_{q,\ell-1}, \qquad \textit{hence} \qquad K_{q,\ell} \ge \left(\frac{q}{3}\right)^\ell. \tag{14}$$

Proof sketch. The exponents in the Syracuse recursion are i.i.d. geometric on $\{1, 2, \ldots\}$, so the exponent gap between two independent copies has law $\Pr[a' - a = s] = 2^{-|s|}/3$, and two copies collide exactly when $Y' = 2^s Y$. Each $H_{\ell-1}(s)$ is a probability times a nonnegative constant, so the sum is termwise nonnegative; dropping the off-diagonal terms ($s \neq 0$) can only decrease the right-hand side, giving the displayed inequality. This is the Fourier dual of Theorem 10.14, with $q/3$ its circle average; what the time-domain form adds is termwise nonnegativity, invisible in the signed spectral balance. Full derivation and the numerical verification (identity checked to 2.3×10^{-13} absolute at worst; ratio never below $q/3$, for $q \in \{3, 5, 7, 9, 11, 13, 15, 21, 25\}$) are in the accompanying repository. □

Since $c_{q,d} < q/3$ for every d, (14) strictly improves Theorem 10.14's constant for every $q \geq 5$ where the latter applies, and it covers the exceptional (Wieferich) primes and composite odd q, both excluded there. At $q = 3$ the rate is $q/3 = 1$: only monotonicity, already known from Theorem 10.13. What is new at $q = 3$ is a reduction: writing $G_{\ell-1}(r) := H_{\ell-1}(2r)$, the increment is

$$K_\ell - K_{\ell-1} = 2\sum_{r \geq 1} 4^{-r} G_{\ell-1}(r) \geq \tfrac{1}{2} G_{\ell-1}(1) = \tfrac{1}{2} T_{\ell-1}(4, 1), \qquad T_\ell(u, v) := 3^\ell \Pr[F' = uF + v \bmod 3^\ell],$$

so the missing uniform margin for $q = 3$ reduces to a single affine shifted collision, involving no cancellation, only positivity. Measured through $\ell = 8$, $T_\ell(4, 1)$ rises slowly through 0.72; the infimum of $T_\ell(u, v)$ over every affine state flattens near 0.191. Two natural certificate constructions for a uniform lower bound on $T_\ell(4, 1)$ (state-by-state induction; a finite-resolution subinvariant) both fail, for reasons recorded in the accompanying repository.

Definition 10.16 (Primitive spectral concentration $SC_{\text{prim}}(\varepsilon)$)**.** $\mu_{\ell,j}$ (the law of $Y = \sum R \bmod 3^\ell$) satisfies $SC_{\text{prim}}(\varepsilon)$ if there is ξ with $3 \nmid \xi$ and $|\hat{\mu}_{\ell,j}(\xi)| \geq 3^{-\varepsilon\ell}$.

Allowing every nonzero frequency would make the condition automatic by Proposition 10.11. The primitive restriction is also required by the diagonal configuration. The same proposition shows that a support hole alone cannot imply $SC_{\text{prim}}(\varepsilon)$ for any fixed $\varepsilon < 1$, or even force a nonzero primitive coefficient.

Definition 10.17 (Chained configuration)**.** A chained configuration of length L has a primitive frequency ξ, an interval I of L consecutive scales, and exponents a_t for $t \in I$ satisfying

$$|a_t - \gamma t| \leq t^{2/3}, \qquad \|\xi 2^{a_t}/3^t\| < \delta, \qquad |a_{t+1} - a_t| \leq C.$$

The constants $\delta < 1/2$ and C do not depend on L.

Conjecture 10.18 (Primitive chained concentration target)**.** *For fixed $c > 0$ and sufficiently small $\varepsilon > 0$, failure at scale (ℓ, c) together with $SC_{\text{prim}}(\varepsilon)$ forces a chained configuration at slope γ_c whose length $L = L_\ell$ satisfies $L/\log \ell \to \infty$.*

The unchained diagonal condition above does not control how the exponents at adjacent scales fit together and is insufficient for this target. The Riesz-product and Halász route supplies good bases on a set whose mass may be exponentially small when the Fourier coefficient is $3^{-\varepsilon\ell}$; with fixed ε, the available intra-scale argument yields only bounded chains. The alternative recursion in j is a convex Pascal mixture and does not keep a concentrating frequency fixed. Neither route proves Conjecture 10.18, which is not used elsewhere in the paper. Proposition 10.12 supplies an exact intermediate quantity: a proof from support data must first produce quantitative fresh-fibre imbalance at consecutive scales, and must then align the resulting primitive frequencies into one chain.

10.3 Proposition C: an exact wall of constants

Proposition 10.19 (Unrestricted Halász deficit). *The Fourier-inversion argument for a support hole gives a coefficient of scale at least $3^{-\ell}$ at some nonzero frequency, but that frequency may be coarse and cannot be used in the diagonal configuration. No primitive lower bound follows from a hole, by Proposition 10.11. Even if a primitive coefficient of size $3^{-\ell}$ is added as a hypothesis, Halász's theorem yields the budget $\sum \|\cdot\|^2 \le 0.549\ell$ against a required ceiling of 0.25ℓ ($\delta = 1/2$, $d = \ell$ generators): a deficit of a factor ≥ 2.2, rising to ≈ 7 at a realistic generator density $\rho \approx 0.3$. In Tao's original 2011 application this closes because $\log q \ll d$ (unbounded ratio); in the Weak Covering Conjecture regime, $\log_2 q/d \ge 1.6$, and the technique does more than degrade: its sign inverts.*

Theorem 10.20 (Jensen identity and the exact Fourier-budget deficit). *Let $Z = \sum_{g\ge1} w_g\, e(U_g)$, U_g i.i.d. uniform, $w_g = p(1-p)^{g-1}$ ($\sum_g w_g = 1$), with $p \ge \frac{1}{2}$. Then, exactly,*

$$\Lambda := \mathbb{E}\big[\log(1/|Z|)\big] = \log(1/p). \tag{15}$$

Proof. By the memoryless self-similarity of the geometric weights ($w_g/(1-w_1) = p(1-p)^{g-2}$ for $g \ge 2$, the same geometric sequence re-indexed), $Z \overset{d}{=} p\, e(U_1) + (1-p)\, Z'$, with Z' an independent copy of Z; since $|Z'| \le 1$ always (a convex combination of unimodular terms), and $p/(1-p) \ge 1$ for $p \ge 1/2$, we have $|Z'| \le p/(1-p)$ whenever $p \ge 1/2$. Writing $c := (1-p)Z'/p$, so $|c| \le 1$, factor $p\, e(U_1) + (1-p)Z' = p\, e(U_1)\big(1 + c\, e(-U_1)\big)$ and take $\log|\cdot|$: the function $z \mapsto \log|1+cz|$ is (the real part of) a holomorphic function of z with no zero inside the disc $|z| < 1/|c| \ni$ the unit circle (as $|c| \le 1$), so by the mean-value property (Jensen's formula with no zeros enclosed) its average over $z = e(U_1)$, U_1 uniform, equals its value at $z = 0$, namely 0. Hence $\mathbb{E}_{U_1} \log|p\, e(U_1) + (1-p)Z'| = \log p$ for every realization of Z', and averaging over Z' gives $\mathbb{E}[\log|Z|] = \log p$, i.e. (15). ☐

Applying Theorem 10.20 with $p = p_c = 1/\gamma_c \approx 0.558$ (the tilted model matching the Weak Covering Conjecture's threshold slope; note $p_c \ge \frac{1}{2}$, as the theorem requires) gives $\Lambda = \log \gamma_c \approx 0.5836 + O(c)$, confirmed by Monte Carlo (8×10^6 samples: 0.58344 ± 0.0005), against the value $\log 3 \approx 1.0986$ that would be needed to exclude a diffuse hole by an annealed ℓ^1 Fourier-budget argument: a deficit factor of ≈ 1.88. The naive threshold for inversion, $p = 1/3$, sits below $\frac{1}{2}$, where Theorem 10.20's identity no longer holds: the same stationary-phase argument gives $\Lambda(p) < \log(1/p)$ strictly for $p < \frac{1}{2}$, so the actual inversion point is $\gamma \approx 3.31$ ($\Lambda \approx 1.032$ at $p = 1/3$, against $\log 3 \approx 1.0986$) rather than the naive $\gamma = 3$, still far above the regime of the conjecture ($j^*(\ell)/\ell \approx 1.2$, Empirical Result 7.2). No member of the wider ℓ^r family ($r \ge 1$) closes the gap either: the closing criterion is the same $\|Z\|_r < 1/3$ for every r, and $\|Z\|_r$ increases with r, so $r = 1$ is already the best choice and $r \to 0$ (the Jensen exponent above) is a strict lower bound for the whole family, verified in the accompanying repository.

Proposition 10.21 (Annealed budget for a diffuse hole). *A primitively spectrally diffuse failure of the Weak Covering Conjecture is compatible with the specific annealed ℓ^1 Fourier budget above, with slack ≈ 1.88. That budget therefore does not exclude such a failure. This branch is the one identified alongside $\beta = 1$ (§9) and the finite-level L^2 growth (Empirical Result 10.4) as a further articulation, within the same second-moment Fourier vocabulary as the L^2 condition, of the same missing ingredient, rather than a methodologically independent fourth or fifth route in its own right. Theorem 10.13 identifies their exact telescoping relation; all three branches enter the covariance sum* $\mathrm{Cov} \propto \sum_\xi S_1(\xi)S_2(\xi)^*$ *of §8.*

Given the fragilities listed above, this is a reduction of the spectrally *concentrated* branch to the Erdős–Lagarias–Furstenberg family, conditional on the open stability step, together with a calculation showing that one annealed ℓ^1 budget does not exclude the diffuse branch. It falls short of a reduction of the Weak Covering Conjecture to the classical Z-number problem: Theorem 10.21 and the discussion above show that stronger claim would be imprecise.

11 Contextualizing the barrier in the wider literature

A directed literature search, going beyond the Collatz-specific literature into the general theory of Fourier decay for self-similar measures, situates the barrier against that wider theory and rules out several plausible-looking shortcuts, each for a specific, verified reason.

11.1 The Syracuse measure is a rigid 3-adic self-similar measure

Unrolling the renewal recursion of (1) gives, in the limit,

$$X = \sum_{k\geq 1} 3^{k-1}\, 2^{-S_k}, \qquad S_k := a_1 + \cdots + a_k, \tag{16}$$

the fixed point of the countable affine iterated function system $\phi_a(x) = 2^{-a}(1+3x)$ on $\mathbb{Z}_3$, weighted by $\Pr(a) = 2^{-a}$, every map sharing the *same* ultrametric contraction ratio $|3 \cdot 2^{-a}|_3 = 1/3$. This is still a self-similar structure, non-standard, countable, and overlapping, so the question of whether the general theory of Fourier decay for self-similar measures [21, 22, 23] transfers is a fair one to ask, and we checked it explicitly.

Proposition 11.1 (Rigidity of $\mathbb{Z}_3^\times$). *Every infinite closed subgroup of $(\mathbb{Z}_3, +)$ has finite index (it equals $3^k\mathbb{Z}_3$ for some $k \geq 0$); consequently, since $\mathbb{Z}_3^\times \cong \mathbb{Z}/2 \times \mathbb{Z}_3$, every infinite closed subgroup of $\mathbb{Z}_3^\times$ has finite index. Since 2 is a topological generator of $\mathbb{Z}_3^\times$ (a primitive root modulo 3^n for every n), the unit phases 2^{-a} spread maximally, the opposite of the "thin multiplicative orbit" phenomenon (a Pisot-type obstruction to the Rajchman property) that drives non-uniform Fourier decay on the real line.*

Theorem 11.2 (The general theory does not transfer). *(a) The pushforward mechanism of Baker–Banaji [21] (Archimedean stationary phase, requiring a C^2 map with $f'' \neq 0$ off a null set) would give, at best,* qualitative *Rajchman decay for the analogue considered here, strictly weaker than the super-polynomial decay already guaranteed by Proposition 1.1.*

(b) Brémont's Pisot-type criterion for non-Rajchman self-similar measures, transported to $\mathbb{Z}_3$, trivializes by Proposition 11.1: there is no room for a "thin" 3-adic multiplicative orbit of the kind the criterion requires.

(c) The renewal-theoretic technique of Li–Sahlsten [23], which needs two distinct *contraction ratios with a Diophantine relation between their logarithms, also degenerates: every branch of (16) has the* same *3-adic contraction ratio 1/3, so the relevant "scale walk" is deterministic ($S_n = n \log 3$, a pure lattice walk with no overshoot fluctuation), in place of the non-lattice random walk the technique is built to exploit.*

Baker–Banaji's general phenomenon (Rajchman measures with no uniform decay rate are generic rather than pathological, in the class of self-similar/self-conformal measures) is a fair piece of context: it shows that the kind of failure exhibited by Theorem 10.21 has real structural precedent in the reference literature, and supports the framing that a proof of $\beta = 1$, or the Weak Covering Conjecture, needs the specific arithmetic input of this problem, rather than generic smoothness. It should not be over-interpreted as evidence *for* the diffuse scenario itself. Baker–Banaji's construction requires fine-tuning of a continuous parameter (a Liouville-type construction, dense in the family but exceptional in measure; typical parameters give polynomial decay [24, 25]), while the Syracuse measure is a rigid arithmetic object with no tunable parameter, and Proposition 11.1 shows the enabling mechanism (fine Diophantine approximation) simply does not exist on the 3-adic side, which cuts against the diffuse scenario as much as for it. Theorem 11.2 is cited with this disanalogy stated explicitly.

11.2 A related fixed-modulus proposal

Chang [27] studies the compressed odd-to-odd orbit through $X_t = \mathbb{1}[n_t \equiv 1 \pmod 4]$ and the last index of each maximal run of $X_t = 1$. Its Map Balance Theorem gives a map-level discrepancy of one between two gap classes. The remaining proposed condition asks for a bound, with an unspecified threshold $\delta_{\max}$, on the imbalance between residues 9 and 25 modulo 32 at those times.

The claimed implication to Collatz is not self-contained in [27]: the threshold is delegated to a block-TV budget in a companion work. The current version of that companion work [28] instead gives a conditional implication from equidistribution at growing moduli, together with uniform control of unbounded gap tails. Fixed-modulus balance does not supply those stated hypotheses. We therefore use the modulo-32 statement only as a related empirical diagnostic, distinct from a reduction or an open problem of this paper.

We tested this condition directly on real orbits. An ensemble of many moderate orbits (2–5×10^4 roots, 20–50 bits) shows the aggregate deviation plateauing at 1.2–1.9%, which is on its own consistent with the conjecture (only boundedness is claimed here, convergence to zero is a separate, stronger question); a more faithful test, a single very long orbit (a random 16000-bit start, 38395 compressed steps, 4761 relevant burst-endings), shows the deviation tracking the $1/\sqrt{i}$ statistical-noise curve closely, with no systematic trend, ending at a deviation of 0.00053. Five further orbits from 500 to 8000 bits show the same pattern. These measurements concern only the fixed-modulus statistic. They do not test growing-modulus equidistribution or uniform integrability of gap lengths.

12 Discussion

Neither candidate lemma of §10 produces a proof. Executing them produces three results instead: Proposition 10.2's exact closed-form persistence of coarse correlation, Empirical Result 10.4's finite-level test of a specific sufficient condition, and Theorem 10.20's exact identity quantifying, to a constant factor (≈ 1.88), why a whole family of Fourier-analytic techniques cannot close the gap. The first and third are proved. The second is evidence through level 17 against a bounded second moment; divergence itself remains unproved.

A related arithmetic statement, some form of equidistribution or near-independence of 3-adic digits along the true, deterministic integer dynamics rather than an idealized i.i.d. model, recurs across four routes: the smoothing-transform/endogeny route of §4; the support identity between Wirsching's Weak Covering Conjecture and Tao's weighted $\beta = 1$ problem in Proposition 9.1; Wirsching's 2003 functional equation and its Fabius density (§9); and the second-moment Fourier analysis underlying both the L^2 condition and the spectral dichotomy of Proposition C (§10), which share the character decomposition $\mathrm{Cov} \propto \sum_\xi S_1(\xi)S_2(\xi)^*$ and the exact multiscale identity of Theorem 10.13. The WCC and $\beta = 1$ formulations share an exact support and cost identity, but differ by geometric multiplicity. The L^2 and spectral-dichotomy branches share the same second-moment character decomposition. A barrier reached from only one of these routes could be an artifact of that route's technique. Reaching it from branching-process fixed points, integer covering, a real functional equation, and second-moment character sums at once is harder to dismiss that way.

A companion paper adds a further, independent measurement: a calibrated finite-window comparison separating Kontorovich–Lagarias's prediction for the $5x+1$ counting exponent from Volkov's competing one, once a raw estimator's own bias is calibrated out with matched synthetic controls (Empirical Result 6.2, [9]). Its validity does not depend on any of the framing above.

13 Conclusion and open directions

We summarize the eight tracked points. O6 is an audit closure; the other seven contain open mathematical statements.

(O1) **Cancellation of centered fine-frequency correlations across sibling subtree aggregates** (Theorem 4.3, Remark 4.4). Pairwise fresh-digit independence is impossible: Theorem 8.2 gives total variation $1 - 3^{-s}$ and mutual information $s \log 3$ for every fixed path pair. Averaging over the branching measure on the sibling gap does not repair this in either total variation or mutual information. O1 remains open at the status established above.

(O2) **Wirsching's Weak Covering Conjecture** (Conjecture 7.1), and the stronger weighted covering estimate needed for Tao's $\beta = 1$ (§9). Proposition 9.1 identifies their common support. Theorem 9.2 proves that even ideal multiplicity inside the WCC cost slice remains exponentially insufficient; the weighted estimate must reach the typical cost window. Theorem 9.3 now gives an unconditional bound, $\beta_{\text{eff}} \leq 1.882712$, from an inter-level identity independent of this microcanonical machinery. A pairwise anti-concentration inequality would close the remaining gap to $\beta_{\text{eff}} \to 1$, though it is not shown to be the only route; the natural 3-adic Weyl-sum approach to it fails for a structural, unitarity reason. Proposition 9.4 shows this pairwise inequality, if true, would itself rule out a Haar-singular limit at $q = 3$, so it may demand more than the Weak Covering Conjecture strictly needs; a finite check through $\ell = 16$ gives no sign of the singular scenario.

(O3) **Wirsching's Conjecture 2** (§9), above the numerically tested Conjecture 3. A companion paper proves Conjecture 1 (Theorem 9.6, [10]). Theorem 9.8 proves that its target condition ($\star$3) implies the weighted cylinder estimate in O2. Proposition 9.9 identifies the conditional local limit theorem needed to pass from canonical weights to fixed cost. Theorem 9.13 shows that their likelihood gap is at most polynomial at every precision, but in the direction opposite to the required lower bound. Reading Wirsching's chain directly against the primary source shows Conjecture 2's actual target, condition ($\star$4), is a statement purely about the one-dimensional averaging operator W_3 carrying no information about the residue coordinate the rest of the chain needs; a sufficient condition targeting it fails for exactly this reason [10].

(O4) **Exponential control of the fine-frequency tail at depth $\ell \asymp D$** (§8, Regime 3), after separating sublinear-conductor modes. Theorem 9.11 proves that those modes are approximated by the corresponding coefficients of μ_r. Fixed-conductor coefficients need not decay. Theorem 9.12 shows that equivalence already fails at linear scale for the underlying cost vectors, although this separation need not survive the residue map. The remaining linear-precision estimate is the input required by the second-moment route; Remark 8.3 records nearby open theories without claiming an equivalence. Posed per first-step class, O4's pointwise target would imply, via the same Parseval identity underlying Theorem 10.13, the L^2 finiteness condition of O7, a bridge in the opposite direction from the one used elsewhere in this paper. A separate finite-level measurement of the primitive spectrum, and of how much mass the sublinear-conductor separation this regime's own argument requires actually removes, is not reported in this paper. O4 does not reduce to O2 regardless: inverting a pointwise Fourier bound over 3^r frequencies costs a factor $3^{r/2}$ above the cylinder scale O2 controls.

(O5) **Exclusion of primitively spectrally diffuse Weak Covering Conjecture failures**. Proposition 10.11 proves that a support hole alone need not produce any primitive Fourier coefficient. Proposition 10.12 identifies primitive Fourier energy exactly with imbalance among the three fresh children of each residue. Projective compatibility and inherited holes still allow zero primitive spectrum. Proposition 10.21 shows that the annealed ℓ^1 budget tested here has insufficient strength. The entire wider family of annealed ℓ^r budgets, $r \geq 1$, has since been tested and none closes the gap: the closing criterion is the same for every r and only gets harder as r grows, so $r = 1$ was already the best member of the family, and the deficit's true inversion point is $\gamma \approx 3.31$, above the naive threshold $\gamma = 3$ used as a first pass earlier, since the identity behind Theorem 10.20 requires $p \geq \frac{1}{2}$ and degrades strictly below it.

(O6) **Fixed-modulus residue balance, audit closed** (§11). The statistic proposed in [27] was tested here, but its claimed implication depends on an unspecified threshold and on a companion framework whose current sufficient hypothesis uses growing moduli and tail control [28]. It is not counted as an open problem of this paper.

(O7) **Absolute continuity and the tail index of the tilted Syracuse density** in the coherent q-adic root environment (Conjecture 3.8), for every odd $q \geq 3$. Theorem 3.7 characterizes its L^p form by normalized collision probabilities. At $p = 2$, Theorem 10.13 identifies bounded collision mass with summability of the primitive energies from O5. The matching identity in Theorem 10.14 gives exponential collision growth for the nonexceptional primes $q \geq 5$ and expresses the remaining $q = 3$ increment as an explicit low-versus-high spectral balance, without yet giving a uniform sign margin there. Theorem 10.15 now gives an unconditional collision rate $K_{q,\ell} \geq (q/3)^\ell$ for every odd $q \geq 3$, prime or composite, with no maximal-lifting hypothesis, strictly improving the constant above for every $q \geq 5$ where it applies and covering the exceptional primes it excludes; at $q = 3$ it reduces the missing margin to a single affine shifted collision, positivity only, no cancellation, measured but not yet uniformly bounded below. The matching i.i.d. statement is Theorem 3.9; its arithmetic transfer remains open. The companion paper's real-tree root batteries bear on a formally separate companion conjecture there, for the growth scale factor of the arithmetic tree itself, expected to share the same exponent, rather than directly on Conjecture 3.8. A sample of 10^5 roots gives a clean GPD threshold-stability plateau at the predicted value across every threshold level tested, a tightly concentrated bias-corrected Hill estimate consistent with it across all headroom levels, and a Vuong test against a truncated lognormal alternative that does not favor the alternative. Two calibration checks (against an exact synthetic Pareto sample of the predicted index, and against a deliberately mis-specified normalization exponent) found no artifact behind this reading. That companion conjecture is strongly supported for $q = 5$, though not proved: the Vuong test yields non-rejection rather than confirmation, and the finite-depth factors remain pre-asymptotic at the headroom reached (their median continues to drift with headroom even as the estimated tail index does not). A direct, exact test of W_q's own tail, the martingale of Conjecture 3.8 itself, remains inconclusive.

(O8) **The exponent transition on the actual arithmetic tree** (Conjecture 3.5): established here only for the matching i.i.d. branching-random-walk model (Theorem 3.3, and, conditionally, Proposition 3.4); for $q = 3$ this is the Growth Exponent Conjecture of Kontorovich–Lagarias and Applegate–Lagarias, open since 1995, of which the best unconditional partial result remains the one-sided bound of Krasikov–Lagarias [15]. A companion paper gives a calibrated empirical measurement at $q = 5$, separated from a competing prediction (Volkov's, discussed in [11]) by seven band-widths once the estimator's own bias is calibrated out with matched synthetic controls [9]; this is evidence rather than proof, and it targets the disputed exponent value rather than Volkov's branching model itself.

Given the structural identities established here, progress on any one of (O1)–(O8) would likely transfer to several of the others at once.

Data Availability Statement

This paper's own claims (§4 onward, excluding the pressure-equation facts restated without proof from §3 and the results restated from §6 and §9's Wirsching subsections, each attributed to its companion paper below) are proved in the text. The following are additionally verified in public scripts, organized by section, at `https://github.com/faculdade/collatz-endogeny`: the residual-coupling experiment of §5, the Weak Covering Conjecture test of §7, the K_ℓ/Jensen computations, the Chang residue-balance test, and the checks for Lemma 4.2, Theorem 9.2, Theorem 8.2, Propositions 10.11 and 10.12, Theorems 10.13 and 10.14, and the finite-level certificates for Theorem 9.3 and Proposition 9.4 (`sec9-worst-cylinder-cascade`) and Theorem 10.15 (`sec10-diagonal-shifted-collision`).

The pressure-equation facts of §3 (including Theorem 3.1, Proposition 3.2, and Theorem 3.9) are proved and verified in full in a companion paper and repository [8]. The Kontorovich–Lagarias vs. Volkov results of §6 (Empirical Results 6.1 and 6.2) are in a companion paper and repository [9]. The Wirsching 2003 results restated in §9 (Theorem 9.6, Theorem 9.8, Theorems 9.10, 9.11, 9.12, and 9.13, and Empirical Results 9.7, 9.14, 9.15, and 9.16) are in a companion paper and repository [10].

Conflict of Interest

The author declares that there is no conflict of interest regarding the publication of this paper.

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A Numerical validation methodology

We record, for completeness, the validation discipline applied to each computational result reported above, since an unvalidated recursive computation is a known source of silent error.

This does not cover the pressure equation, the Kontorovich–Lagarias vs. Volkov comparison, or the Wirsching 2003 material restated without proof in §3, §6, and §9: their own validation methodology is recorded in the corresponding companion paper's appendix [8, 9, 10].

- *Weak Covering Conjecture DP* (§7): validated against an independently computed reference table for $\ell = 1, \ldots, 7$ and a hand-checked point ($\ell = 2, j = 2$).
- *WCC large-deviation barrier* (Theorem 9.2): the exact negative-binomial tail was evaluated in log space through $\ell = 1000$ and compared with the closed-form rate $I(1 + \log_4 3)$. The computation is a check on the constants; the proof uses the exact mass formula and Stirling's formula.
- *Primitive Fourier hole barrier* (Proposition 10.11): direct character sums through $\ell = 8$ reproduce $1/(2 \cdot 3^{\ell-1} - 1)$ for every primitive coefficient of the one-hole unit law. They also give zero, up to floating-point error below 6×10^{-13}, for every primitive coefficient of a non-full law lifted uniformly from the preceding modulus.
- *Primitive fibre energy* (Proposition 10.12): random integer count vectors through $\ell = 8$ satisfy (10) with absolute discrepancy below 3×10^{-16}. Uniform lifts give zero primitive energy at every tested level. The computation also checks the sharp integer-count bound $1/T$.
- *Multiscale Parseval decomposition* (Theorem 10.13): the increment identity and its telescoping sum were checked through $\ell = 12$ for the Syracuse laws and for an independent compatible sequence of Dirichlet refinements. The largest absolute discrepancy was below 6×10^{-14}.
- *Logarithmic collision renormalization* (Theorem 10.14): the collision mass from the original Syracuse recursion was compared with the independently evaluated spectral multiplier through $\ell = 12$ at $q = 3$, and through $\ell = 5$ at $q = 3, 5, 7, 11, 13, 17, 19$. These primes include several multiplicative orders and cases where 2 is not a primitive root. The largest absolute discrepancy in the general test was below 2×10^{-12} for collision masses as large as 10^4; the logarithmic ordering was checked to be a permutation at every level.
- *Fresh-digit affine coupling* (Theorem 8.2): exact enumeration for sibling gaps $2, 4, 8$, coarse precisions $1, 2, 3$, and fresh blocks through six digits reproduces $1 - 3^{-s}$ in total variation and $s \log 3$ in mutual information. This also checks the orientation of the affine residue map and its carries.
- *K_ℓ recursion* (Empirical Result 10.4): the only initially available check was the hard-coded base case ($K_1 = 5/3$), which does not exercise the recursive machinery; we additionally validated the full distribution μ_ℓ, bin by bin, against direct Monte Carlo sampling of the primary definition (1) at $\ell = 3, 4$, within Monte Carlo noise.
- *L^p collision spectrum* (Empirical Result 10.5): its $p = 2$ column was compared level by level with the independently maintained K_ℓ recursion, agreeing to floating-point precision.
- *Worst-cylinder cascade factor* (Theorem 9.3): the inter-level ratio $R_\ell(u)$ and $\min_u R_\ell(u)$ were computed in exact rational arithmetic through $\ell = 10$, certifying $\beta_{\text{eff}} \leq 1.882712$; the monotonicity of $\min_u R_\ell(u)$ itself has a full proof, cross-checked numerically at these levels.
- *Pair inequality forces non-collapse* (Proposition 9.4): the implication is proved; $\mathbb{E}[-\log N_\ell(U)]$ and $\Pr(N_\ell(U) \leq x)$ for fixed x were computed over every unit mod 3^ℓ through $\ell = 16$, giving no sign of the singular scenario the proposition excludes conditionally.

- *Diagonal shifted collision* (Theorem 10.15): the diagonal-term identity and the bound $K_{q,\ell} \geq (q/3)^\ell$ were checked to 2.3×10^{-13} absolute at worst for $q \in \{3, 5, 7, 9, 11, 13, 15, 21, 25\}$, alongside the proof, which does not depend on this numerical check.
- *Jensen-identity Monte Carlo* (Theorem 10.20): the identity is exact and independent of simulation; the Monte Carlo estimate (8×10^6 samples) serves only as a sanity check on the analytic value, and matches it to four significant figures.
- *Chang residue-balance test* (§11): validated by confirming that no burst-ending event ever produced a residue outside the two predicted values ($9, 25 \bmod 32$) in either the ensemble or single-orbit experiments, matching the source paper's structural proposition exactly.

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